

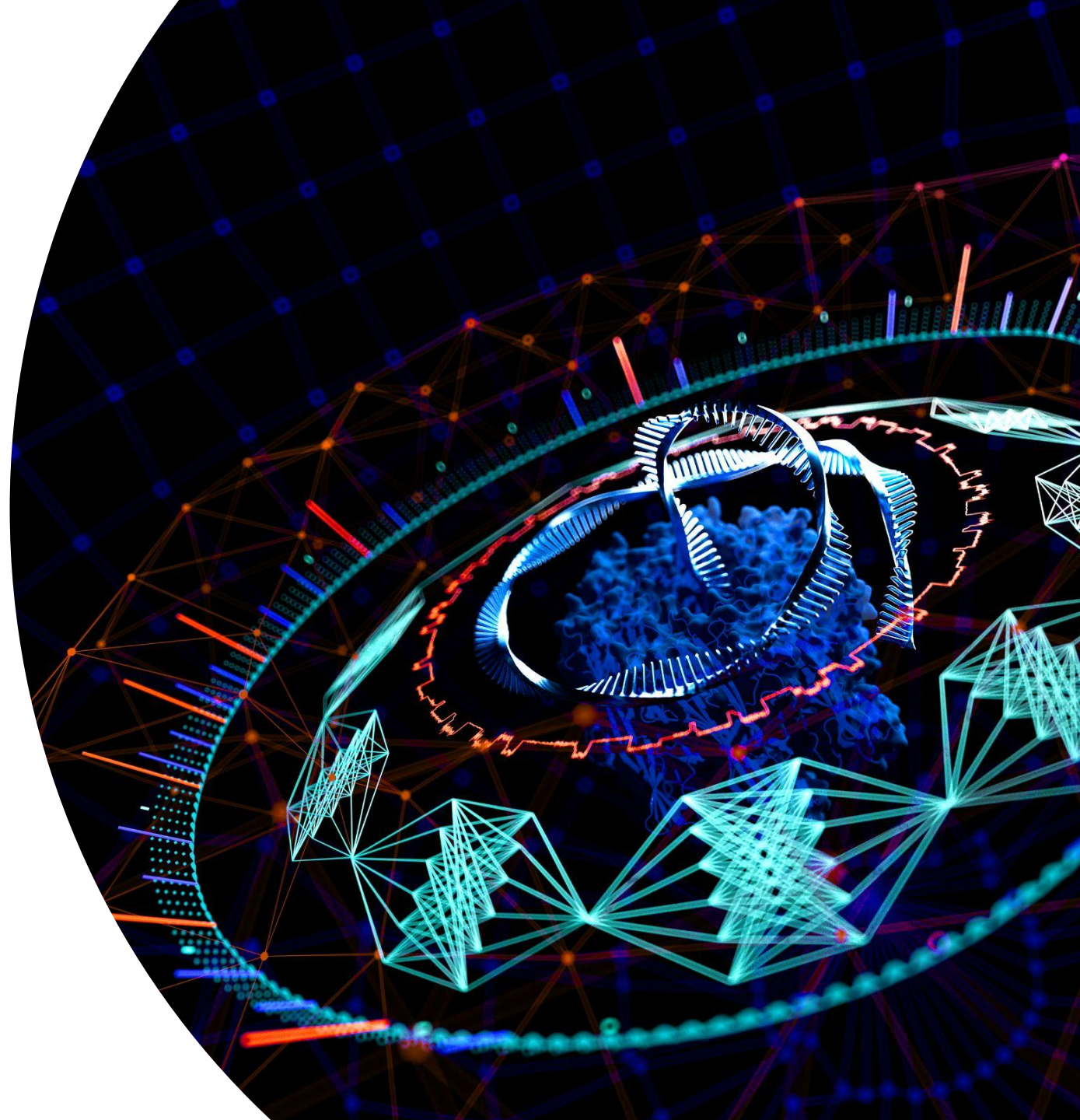
Team project: Identification of RNA modifications from direct RNA-Seq data

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14/9/2026







'Everything is going on normally'- Wuhan's curious nonchalance to virus outbreak

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The inner courtyard of the Wuhan Medical Treatment Center.

Wuhan pneumonia outbreak: China rules out Sars as source; WHO says no evidence of human-to-human

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The viral disease outbreak in Wuhan is still unknown.

Wuhan pneumonia outbreak may be linked to new type of coronavirus; Mers: WHO

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The Wuhan Medical Treatment Center. PHOTO: SCREENGRAB FROM SINA WEIBO

Singapore confirms first case of Wuhan virus; second case likely

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Guests at Shangri-La's Rasa Sentosa Resort & Spa in Singapore had stayed at the resort. PHOTO: TIMOTHY DAVID

Coronavirus: Assemblies, large group activities in S'pore schools to be suspended

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Assemblies, camps, mass celebrations, and other large group and communal activities will be suspended at schools from Feb 5, 2020.

PHOTO: ST FILE



Singapore, April 2020



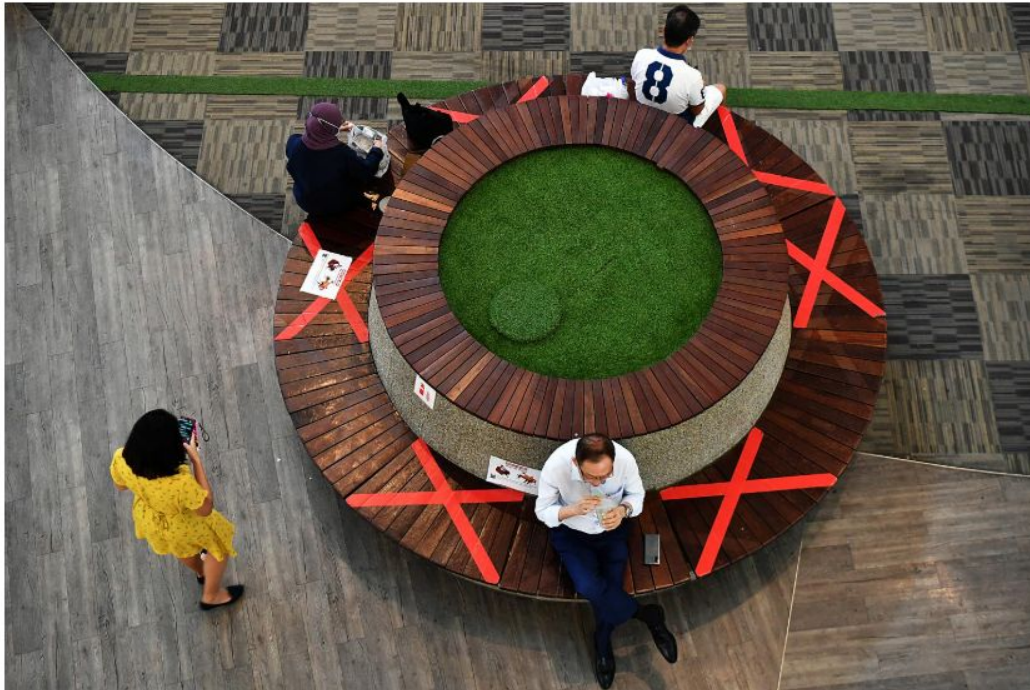
Singapore, April 2020



Jan 2020: the Genome of SARS-CoV2

How a team at A*Star played a big role in global fight against Covid-19 pandemic

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A team from Singapore's A*Star played a major role in setting up and maintaining the database to share and analyse the Covid-19 genomes.

ST PHOTO: LIM YAOHUI

- The first five fully sequenced SARS-CoV2 virus genomes were made public by laboratories in China via the Global Initiative on Sharing All Influenza Data (Gisaid) on **Jan 10, 2020**
- **PCR diagnostic tests** that detect the SARS-Cov2 genome are developed
- **The development of RNA vaccines is initiated**
- Since then, more than **15 million genomes** have been shared with major contributions from Singapore scientists at A*STAR

July 2020: Operation Stronghold



The Stronghold Diagnostics Lab used for Covid-19 testing can be deployed for future screening efforts.

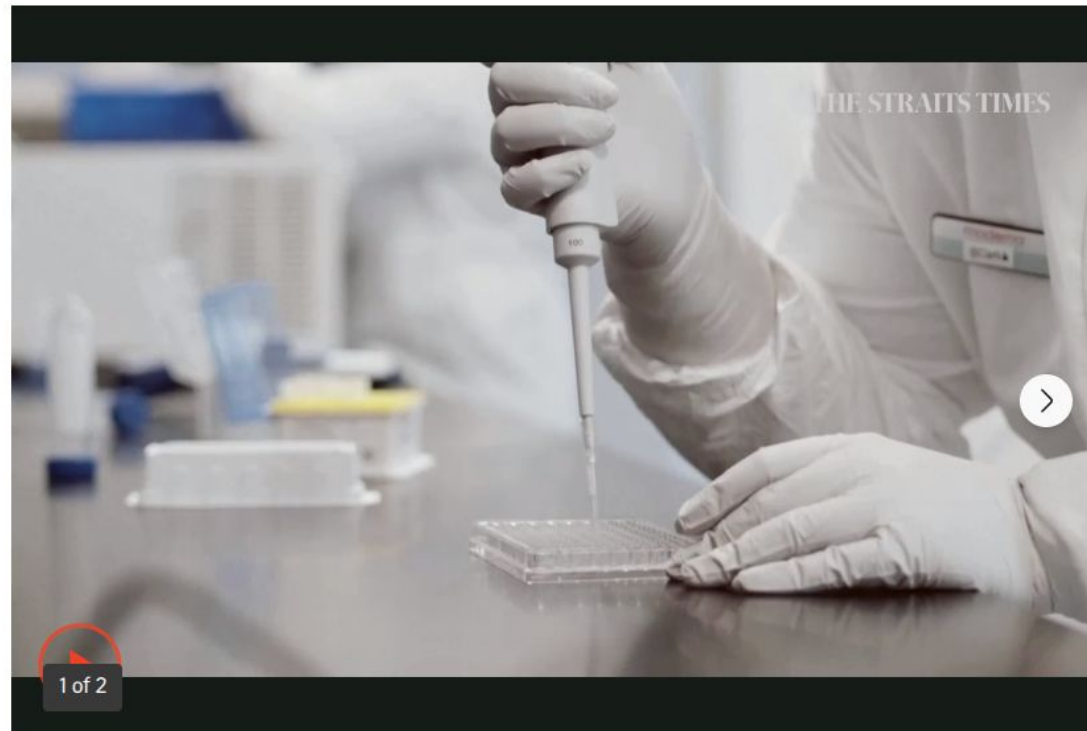
PHOTO: A*STAR

- The Stronghold Diagnostics Laboratories (SDL) is established at the **Genome Institute for Singapore** (A*STAR) for rapid testing using PCR to diagnose SARS-CoV2 infections
- SDL processed **1 million samples** in 2 years

Nov 2020: the first Covid19 vaccines

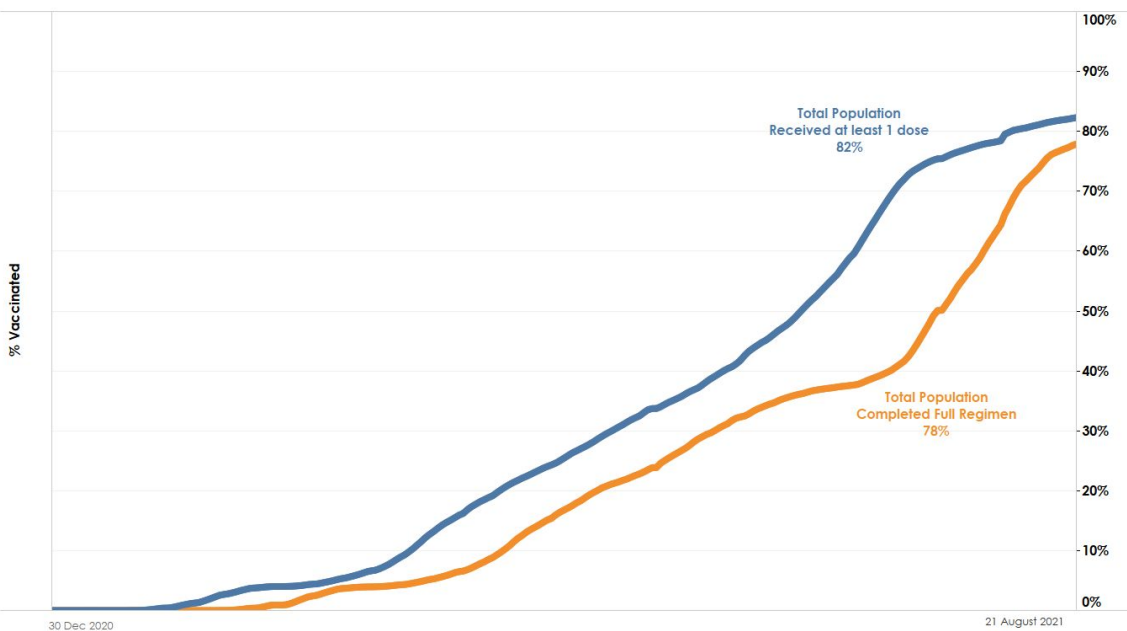
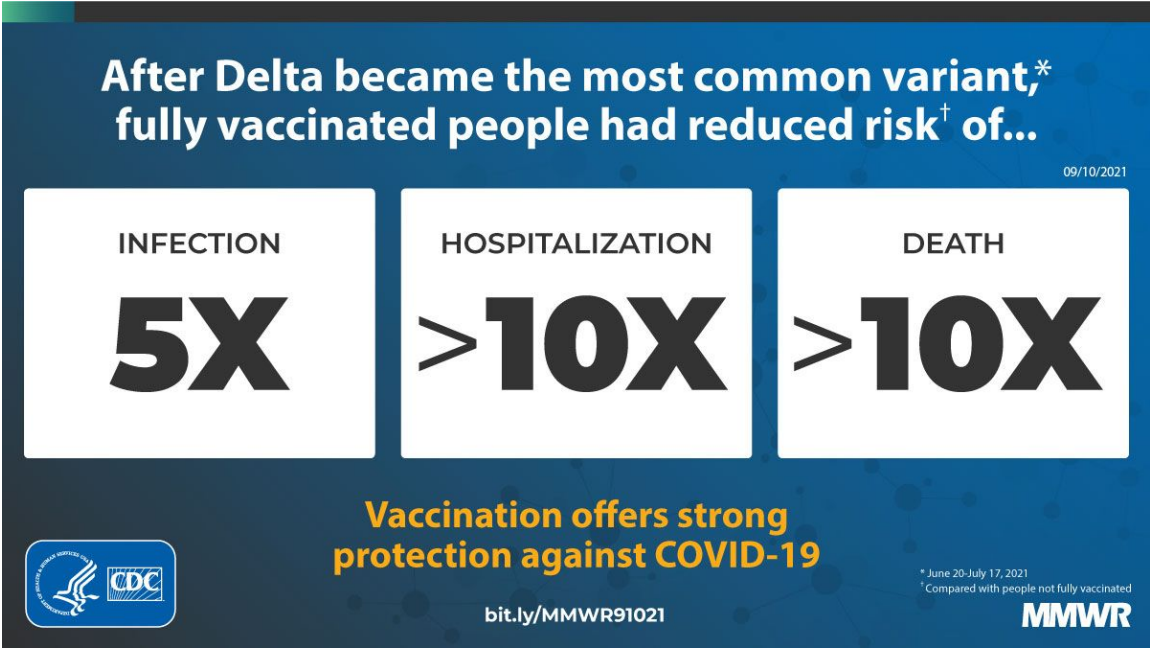
Pfizer ends Covid-19 trial with 95% efficacy, to seek emergency-use authorisation

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Final results from Pfizer Inc's Covid-19 vaccine trial showed its shot had a 95% success rate and two months of safety data.

mRNA vaccines and the end of the COVID19 pandemic



Singapore, 2023

Changi Airport's June passenger traffic crosses 5 million mark for first time since January 2020



June 2023 saw 5.12 million passengers passing through Changi Airport. PHOTO: LIANHE ZAOBAO

The science behind RNA vaccines

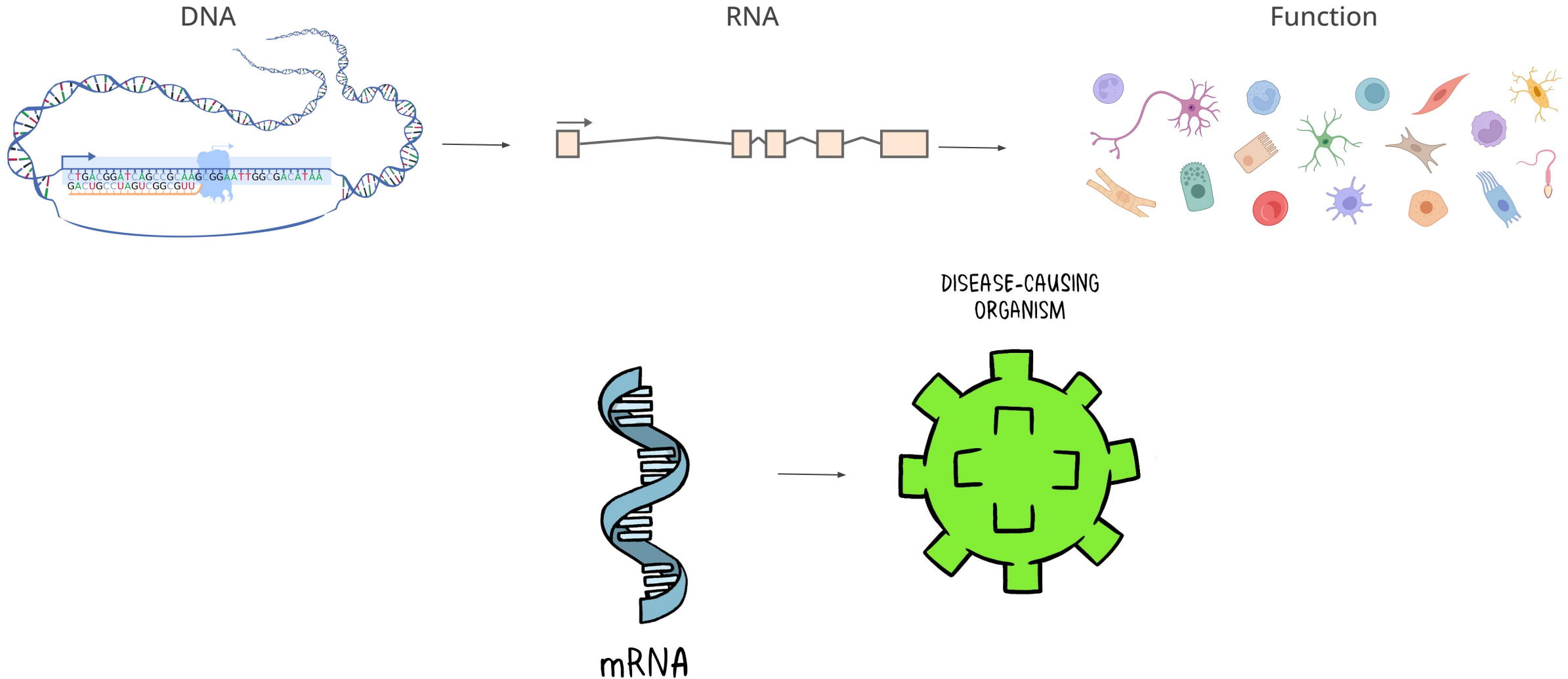
VACCINES

Scientists behind mRNA COVID Vaccines Win 2023 Nobel Prize in Physiology or Medicine

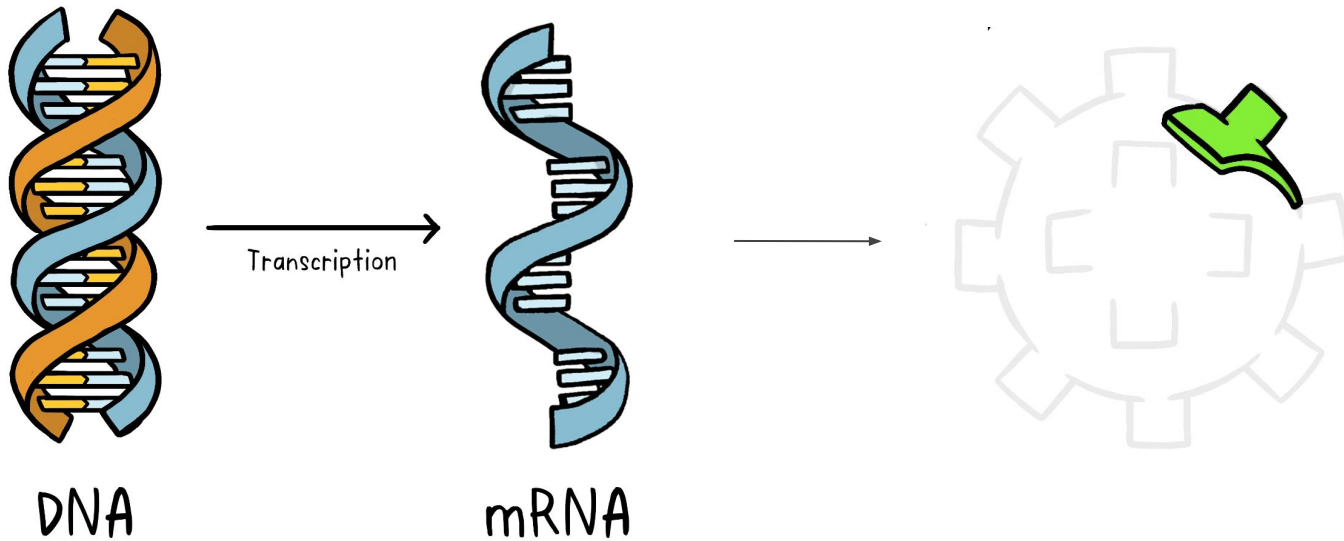
Katalin Karikó and Drew Weissman were awarded this year's Nobel Prize in Physiology or Medicine for mRNA vaccine discoveries that made highly effective COVID vaccines possible

By Lauren J. Young on October 2, 2023

The role of RNA as a messenger



The idea behind mRNA vaccines



- A mRNA vaccine delivers a specific set of instructions (RNA) to our cells
- The RNA is used as a template to make a specific protein is recognized by the immune system

The idea behind mRNA vaccines

mRNA vaccines: challenges

Research on nucleic acid vaccines started in the 1990s against big scepticism:

- How **produce** RNA?
- How to **deliver** mRNA vaccines?
- In vitro transcribed (synthetic) mRNA led to **inflammatory reaction**

-> The RNA itself was targeted by the immune system

Katalin Karikó and Drew Weissman



III. Niklas Elmehed © Nobel Prize
Outreach

Katalin Karikó

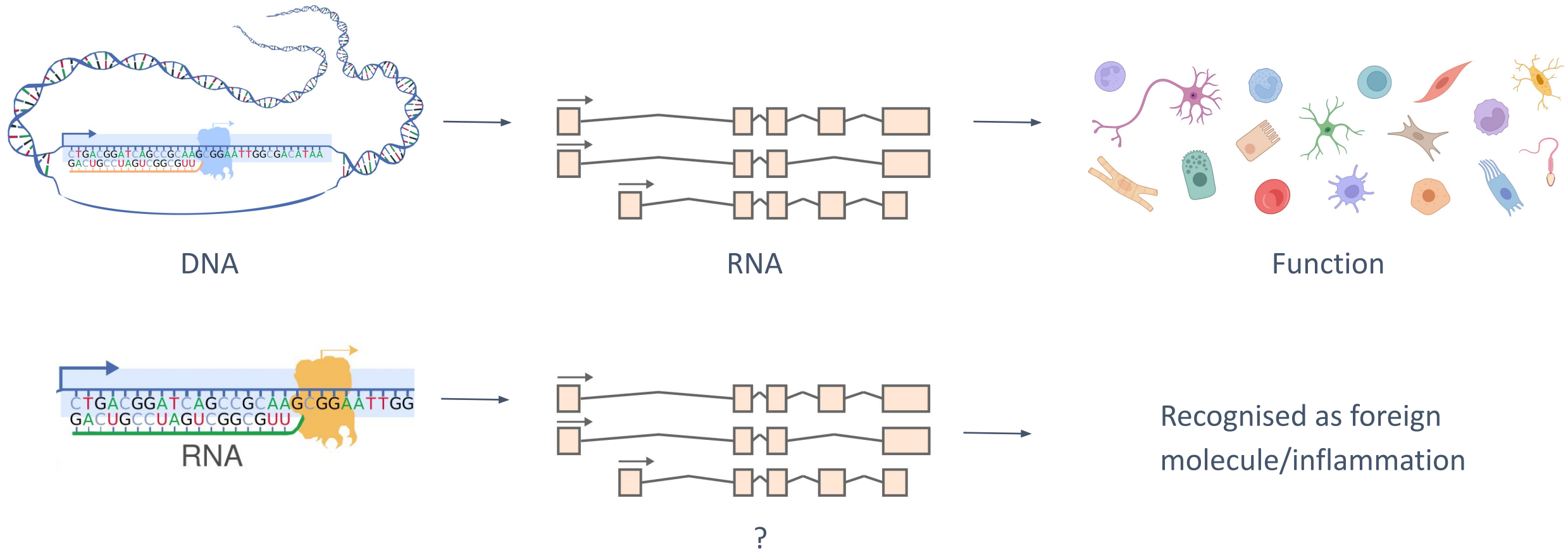


III. Niklas Elmehed © Nobel Prize
Outreach

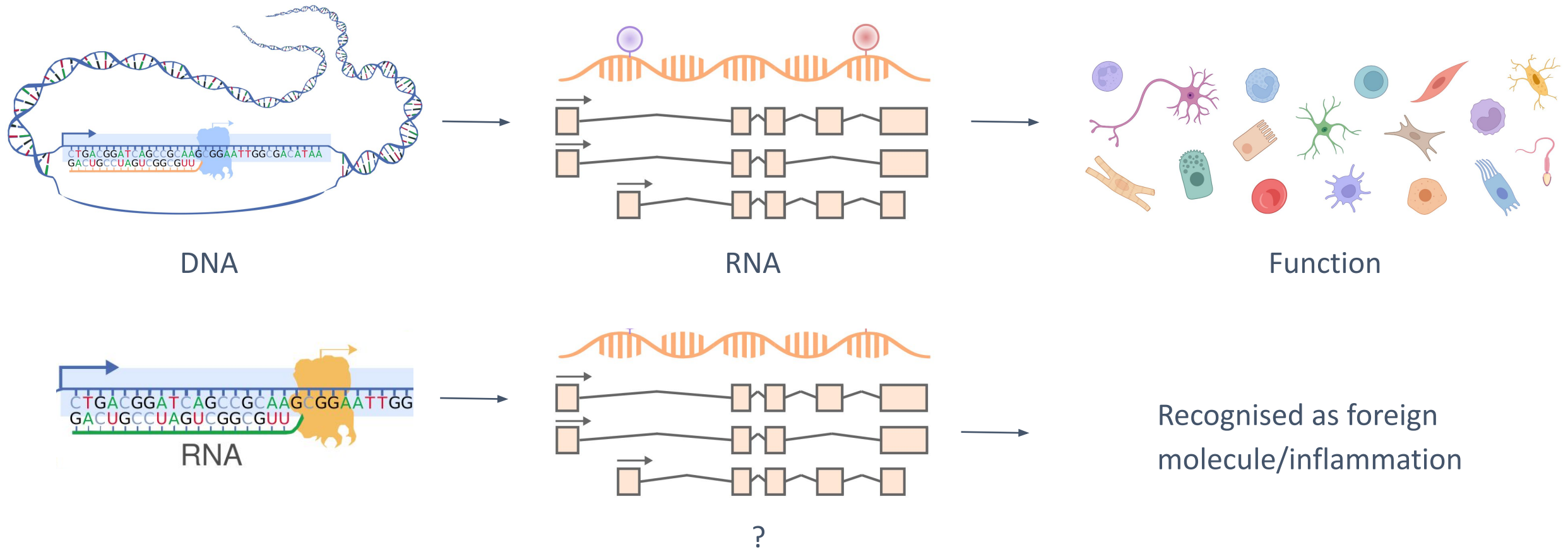
Drew Weissman

- Katalin Karikó and Drew Weissman noticed in **vitro transcribed mRNA is recognised as a foreign substance**, which leads to the release of inflammatory signaling molecules.
- Why is in vitro transcribed mRNA recognized as foreign while mRNA from mammalian cells did not give rise to the same reaction?

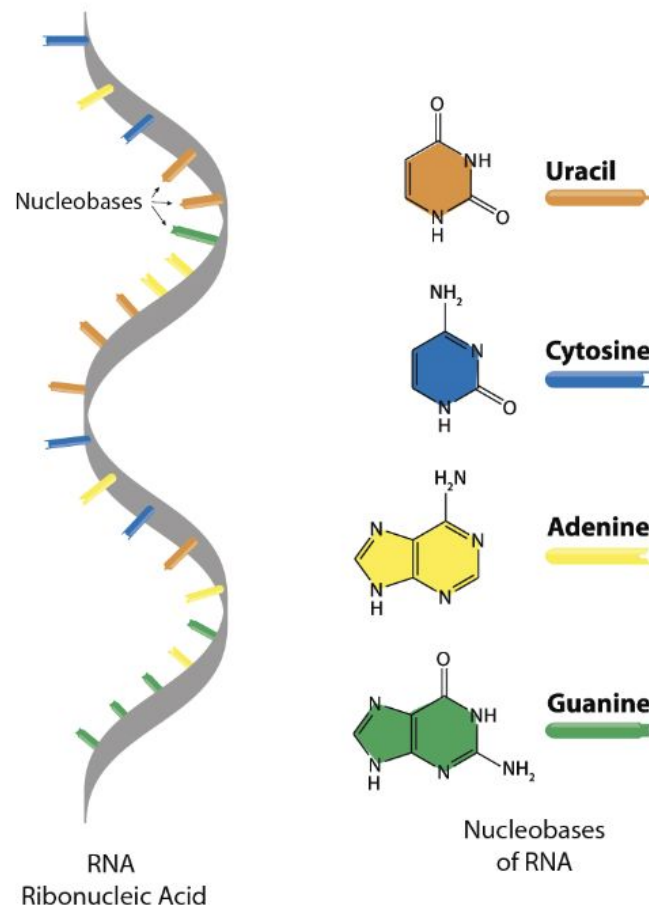
In vitro transcribed RNA: what was different?



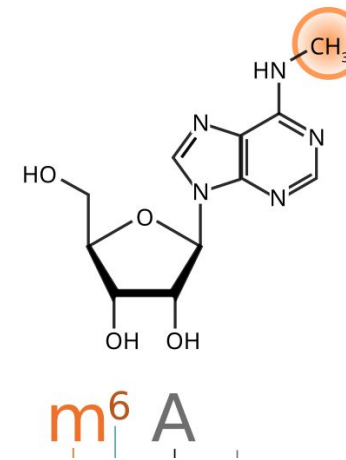
In vitro transcribed RNA: what was different?



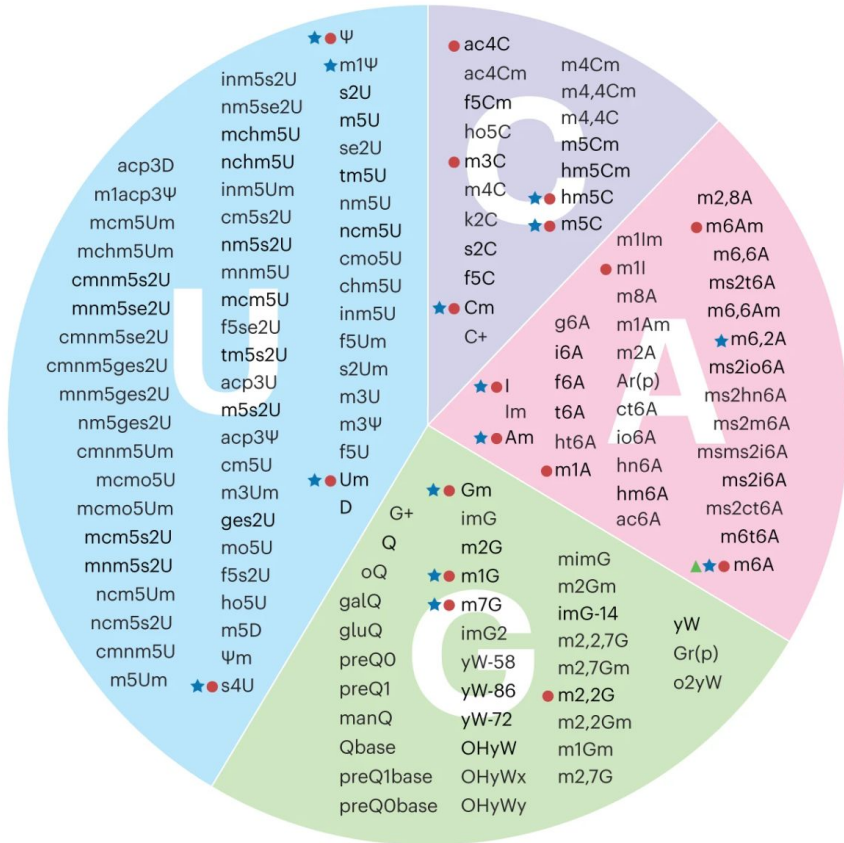
RNA and RNA modifications



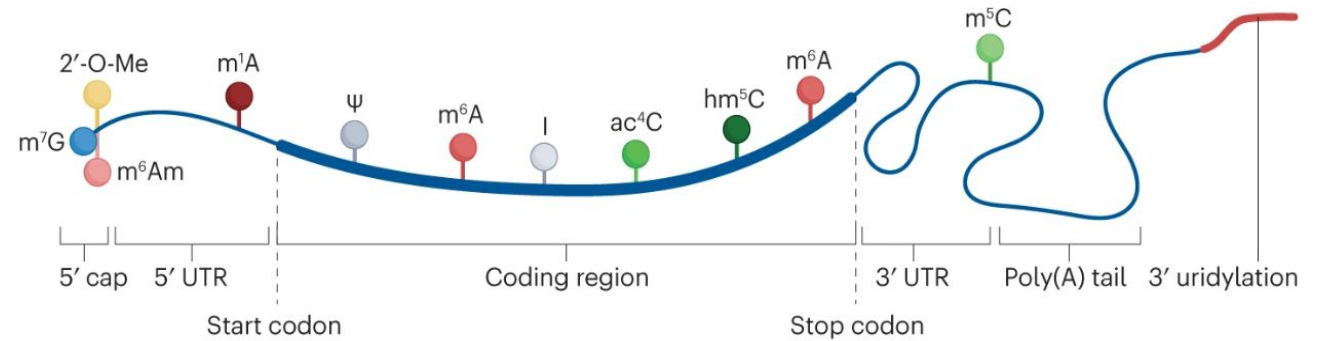
- RNA modifications are chemical modifications of the RNA bases (A,C,G,U)
- The set of all RNA modifications is sometimes referred to as the “Epitranscriptome”



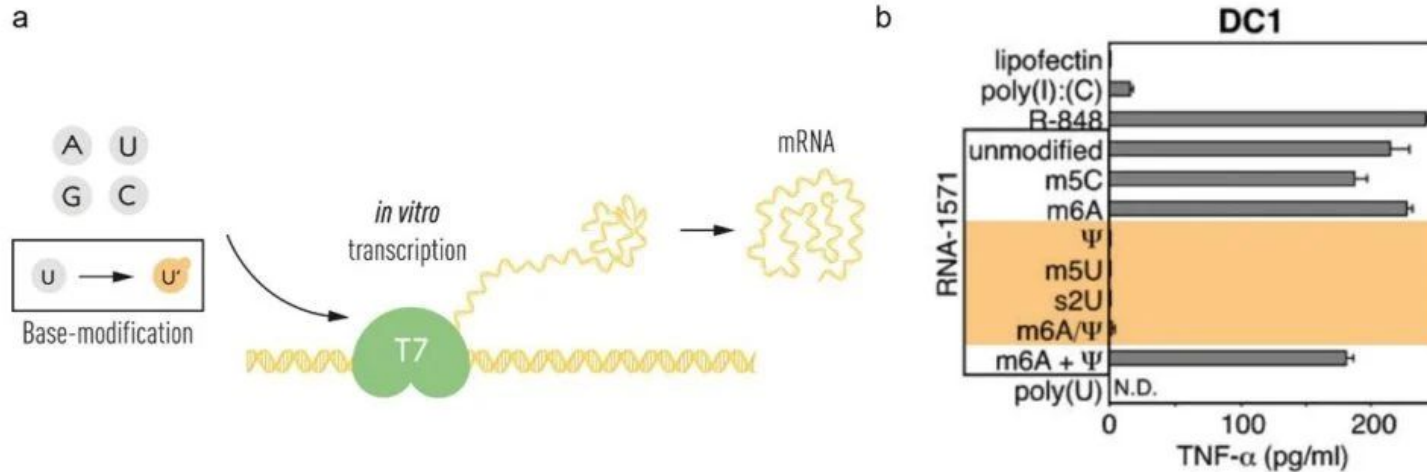
mRNA modifications



- >170 RNA modifications have been described
- mRNAs are frequently modified

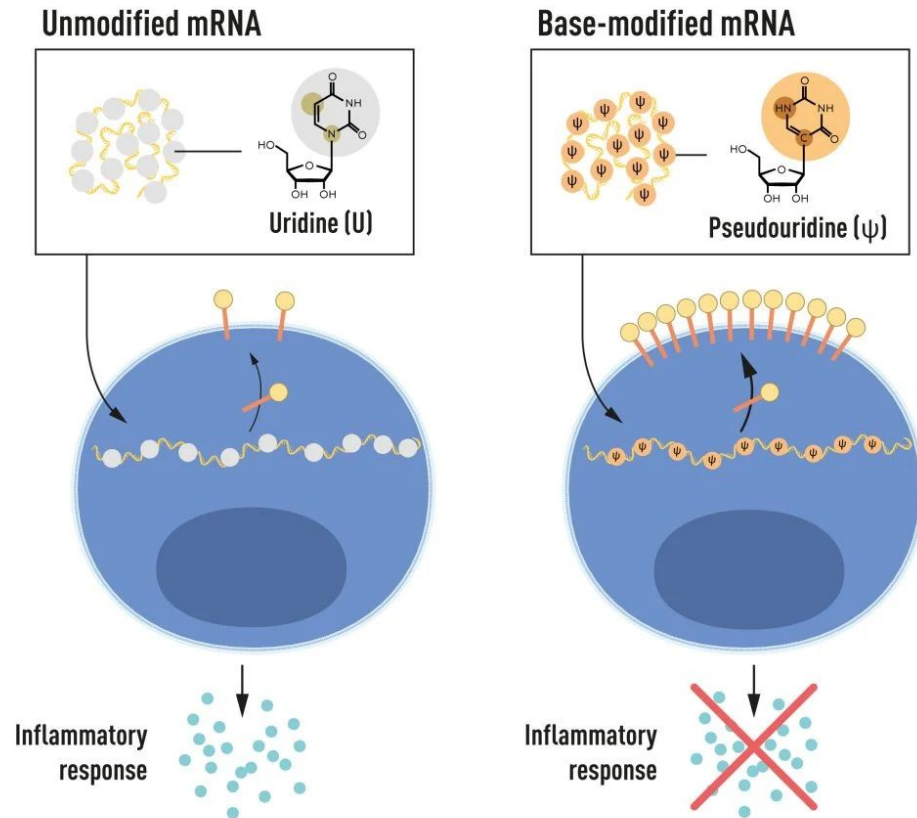


The breakthrough: RNA base modifications



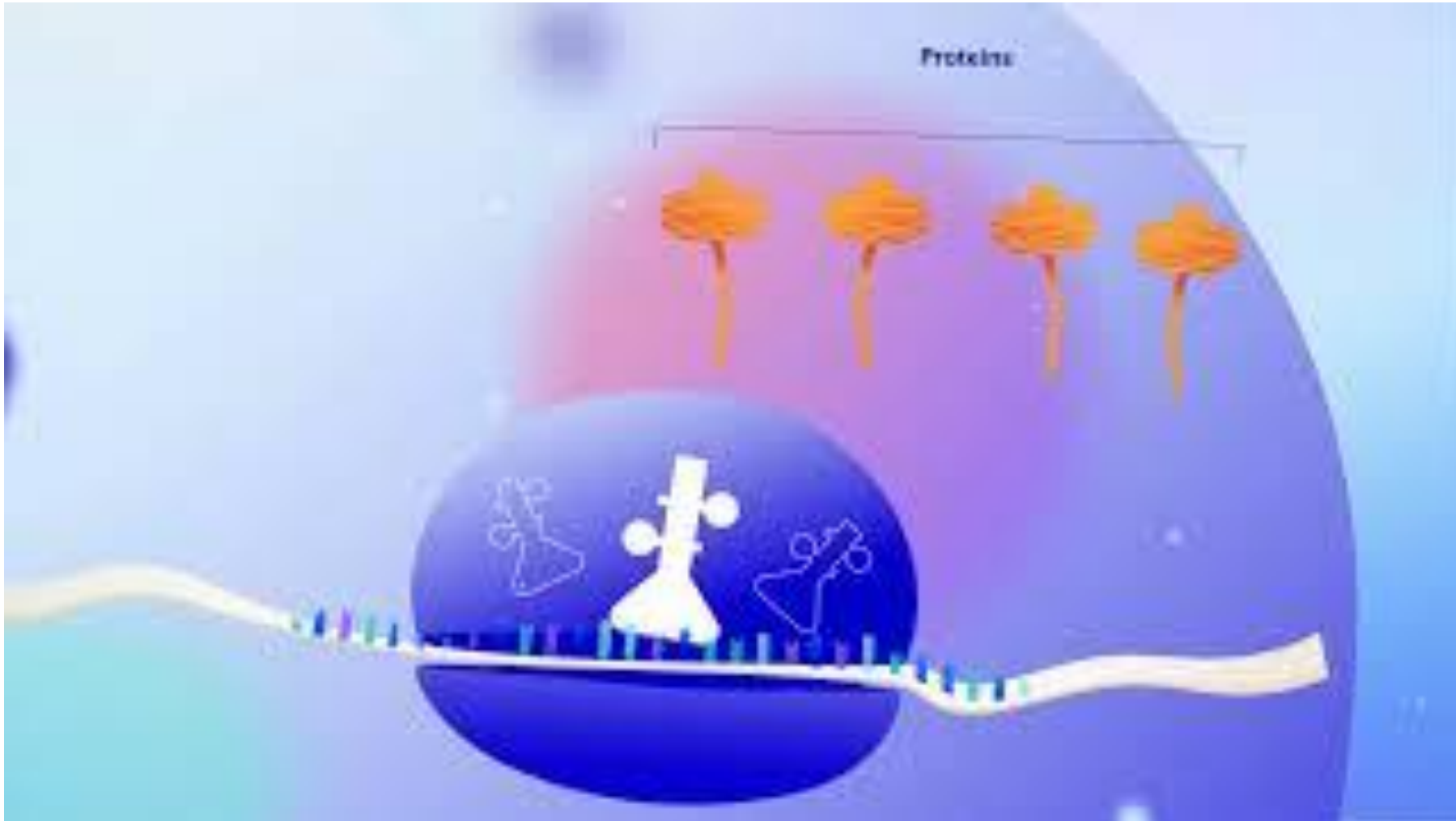
- The use of **base modified RNAs reduced the inflammatory response** (innate immune response)
- N1-methylpseudo-uridine (m1ψ) showed the **strongest reduction in innate immune response** and **higher protein production**

Base-modified mRNA vaccines

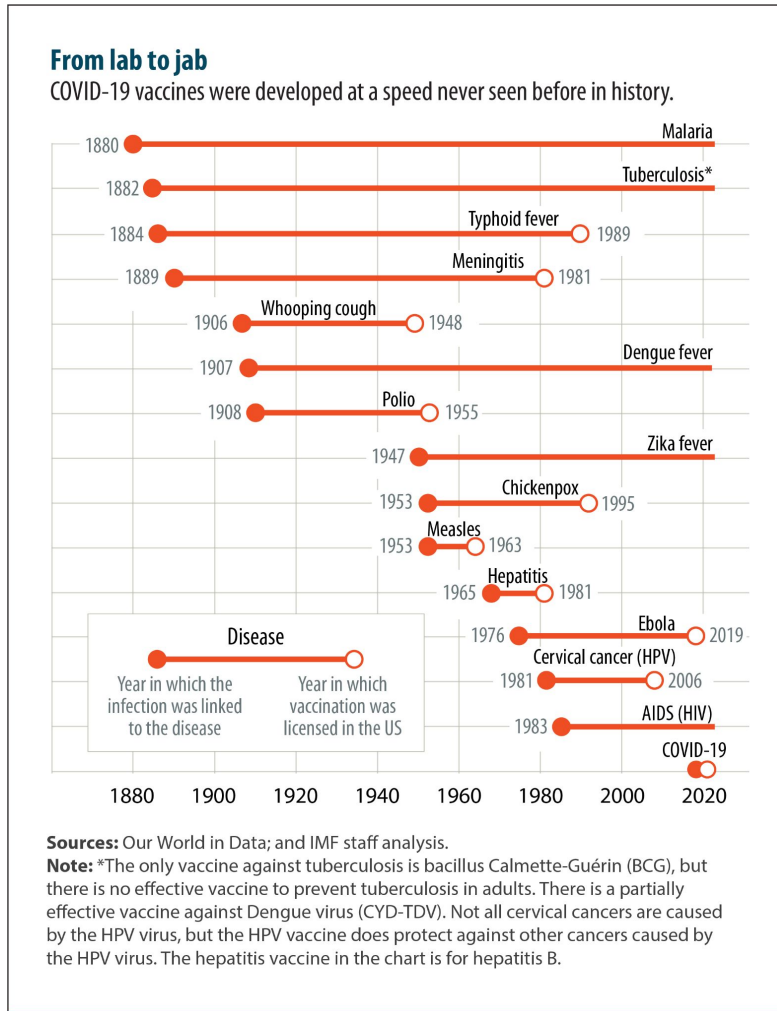


- All approved mRNA COVID19 vaccines use **base-modified mRNA** based on the research by Katalin Karikó and Drew Weissman

mRNA vaccines for COVID19

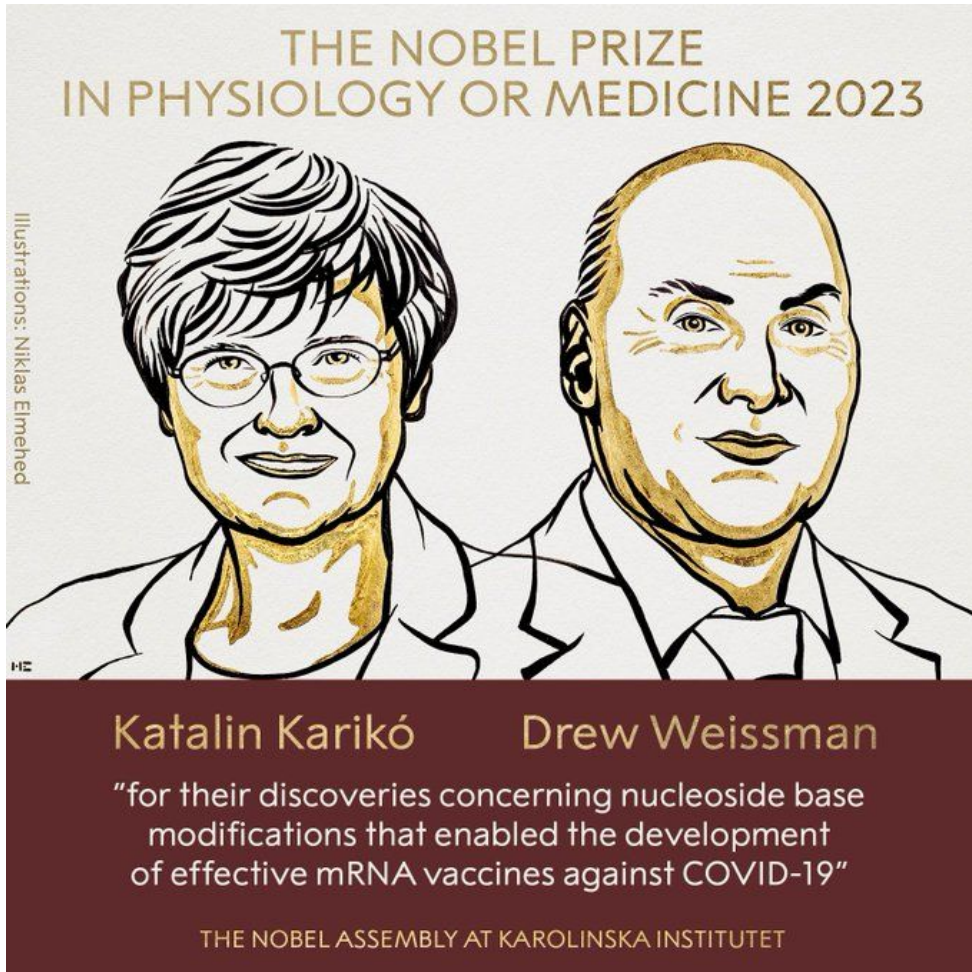


mRNA vaccines for COVID19



- COVID19 vaccines were rapidly developed, approved, and produced
- <1 year between the first genome sequence and the first vaccine approval
- Tens of millions of lives are saved by mRNA vaccines

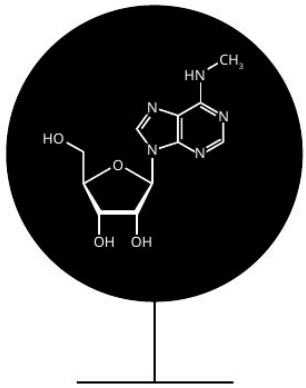
The Nobel Prize in Physiology or Medicine 2023



- Katalin Karikó and Drew Weissman
- For the discovery of **RNA modifications** that enable the **mRNA vaccine** development against COVID19

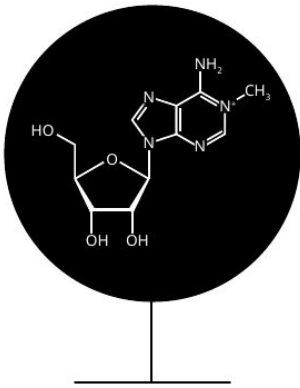
Team Project: RNA modifications

RNA modifications: an invisible layer that impacts RNA function



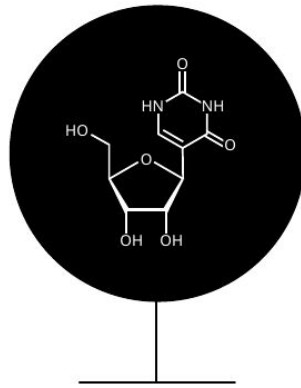
m⁶A

Full name: N6-methyladenosine
Location: mRNA, tRNA, rRNA, snRNA, lncRNA
Known: RNA stability, splicing, export, translation efficiency



m¹A

Full name: 1-methyladenosine
Location: mRNA, tRNA
Known: RNA structure, translation efficiency



ψ

Full name: Pseudouridine
Location: mRNA, tRNA, rRNA
Known: RNA structure, translation efficiency

RNA modifications

- are **chemical alterations** of RNA nucleobases
- **> 170 RNA modifications** have been described

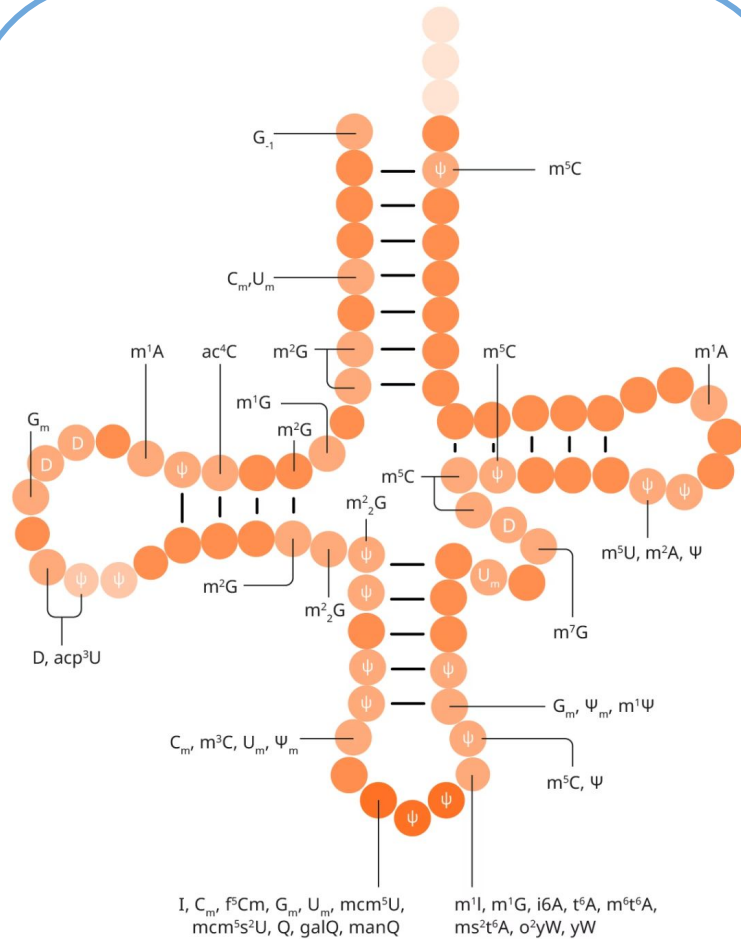
Regulation

- **writer, eraser, and reader enzymes** write, remove, and interpret RNA modifications
- RNA modifications can be **dynamically** modified

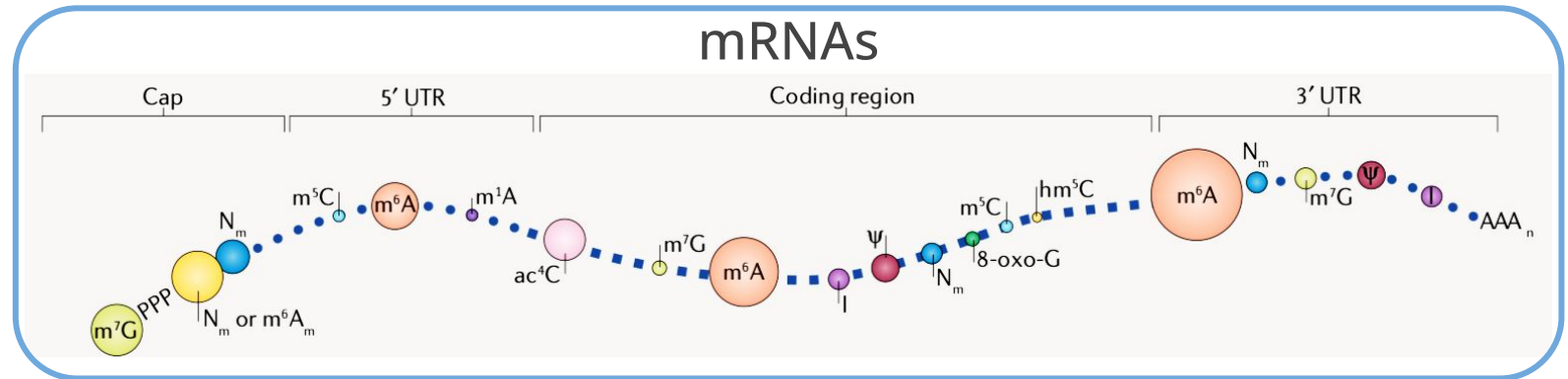
Function

- RNA modifications impact structure, stability, processing, splicing, translation, sequence, immunogenicity
- they are essential for RNA folding, embryonic development, sex determination, mRNA vaccines

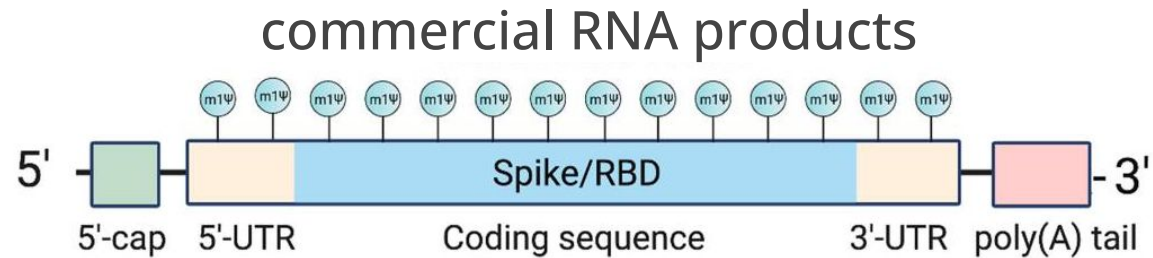
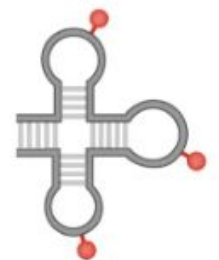
RNA modifications



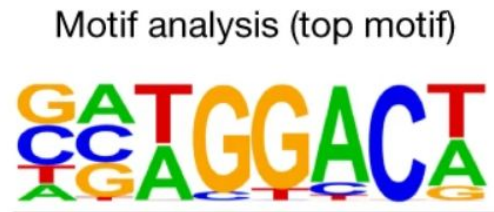
noncoding RNAs



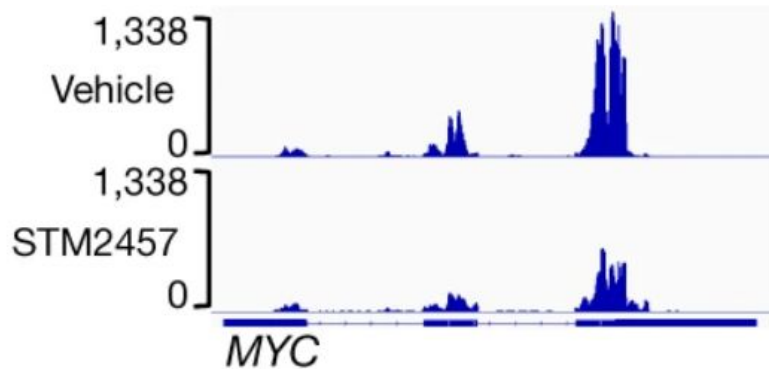
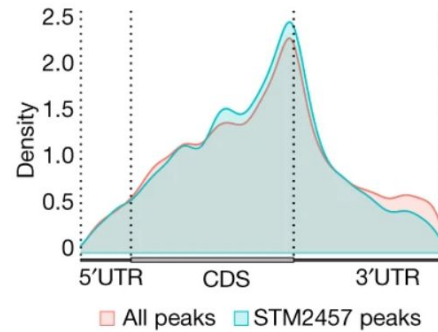
Genomics assays: RNA structure probing, metabolic labeling



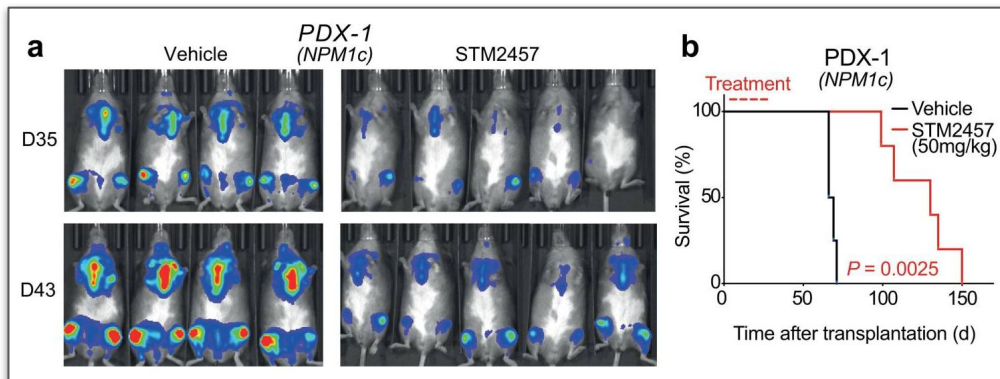
N6-methyladenosine: m6A



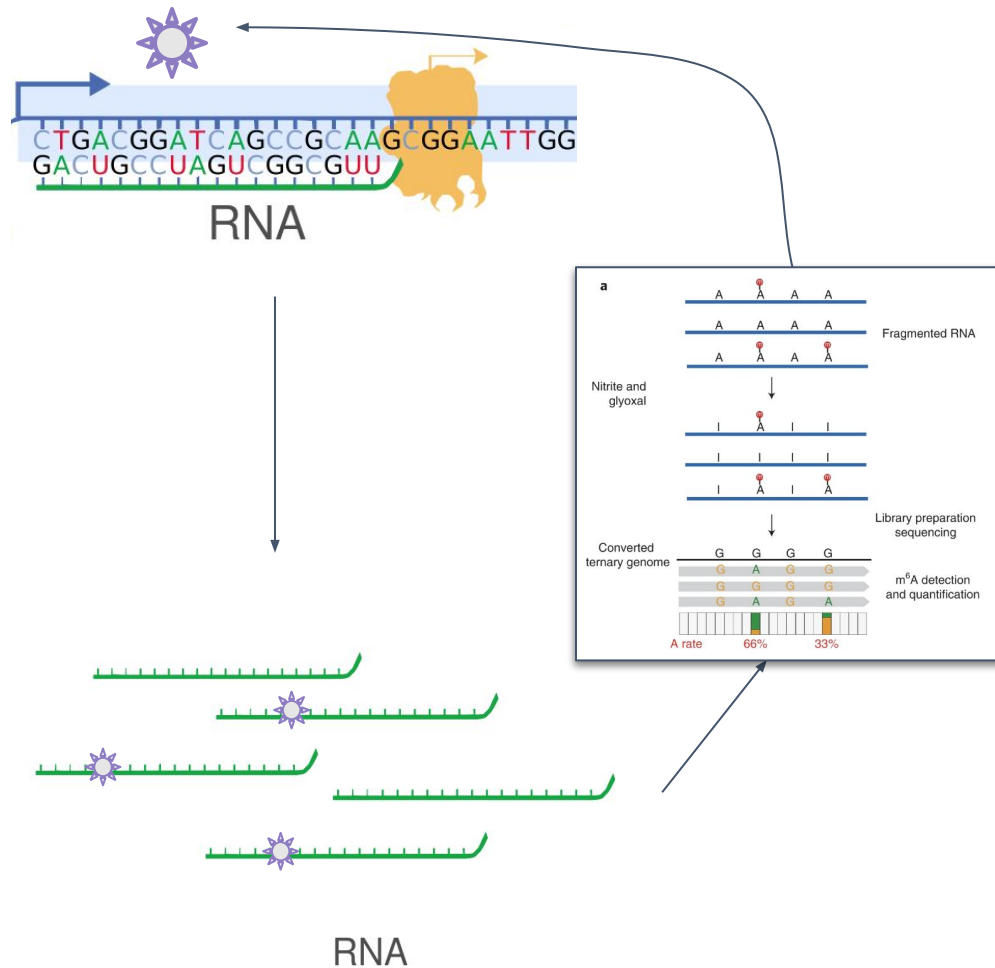
E-value = 10^{-14}



- Writer enzymes: **METTL3/METTL14**
- Binding motif: **DRACH**/GGACT
- occurs at the **3'end** of CDS
- can be profiled genome-wide
- plays a role in acute myeloid leukemia (AML)
- Inhibition of METTL3 leads to reduced AML growth



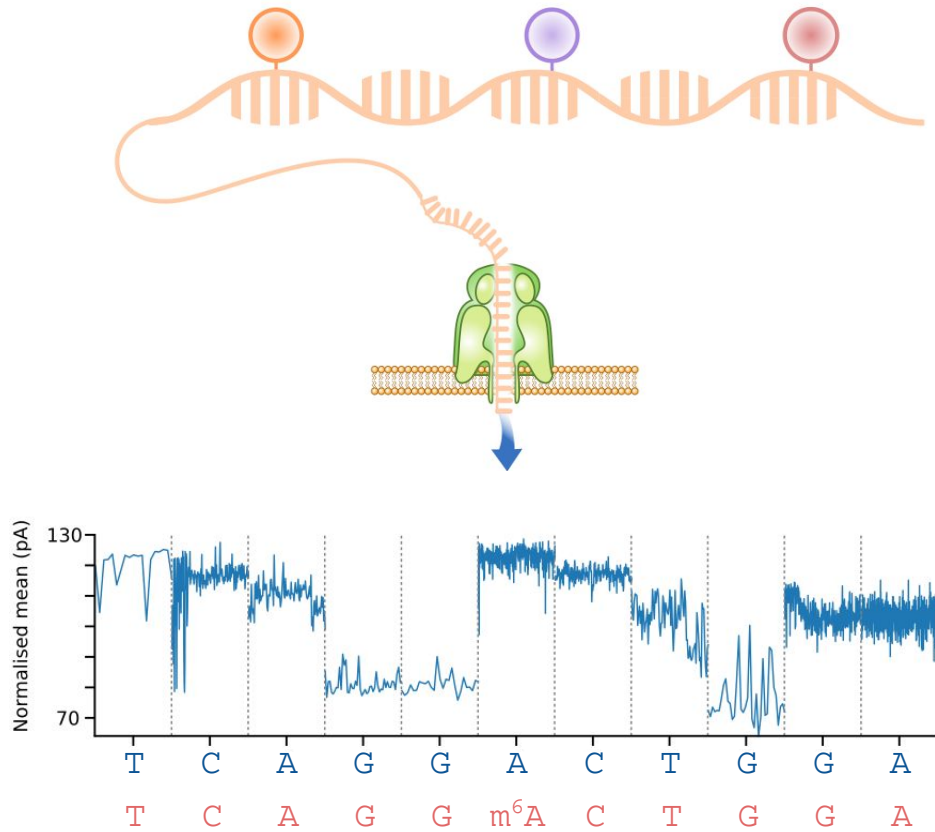
Transcriptome-wide profiling of RNA modifications



Challenges: transcriptome-wide modification profiling

- **specialised protocols**, expensive, time consuming
- **not scalable**: require large amount of RNA input
- only available for **very few modifications**

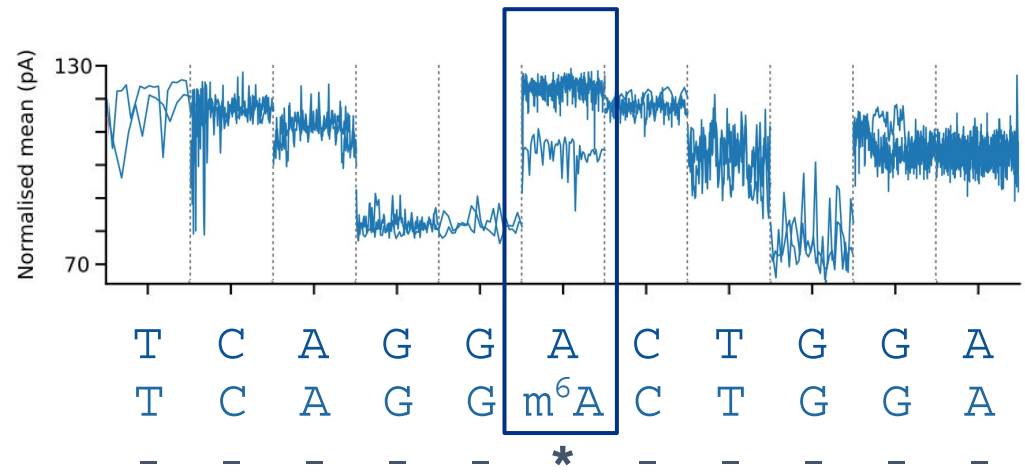
Transcriptome-wide modification profiling using direct RNA-Seq



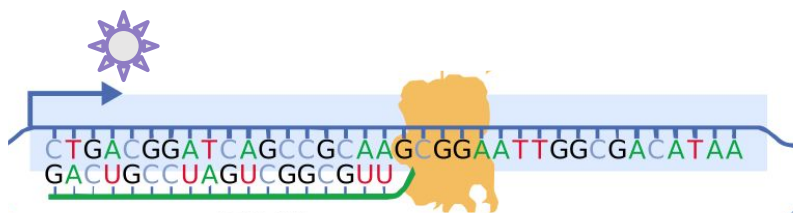
Direct RNA sequencing:

- captures RNA modifications
- single base, single molecule resolution
- transcriptome-wide
- high throughput, easy and cost efficient

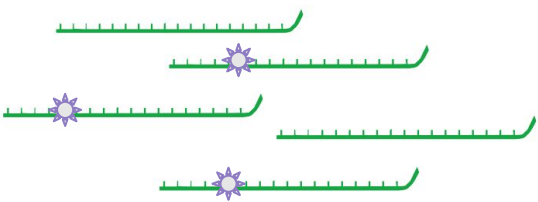
Identification of m6A from direct RNA-Seq data



Labels: Site-level



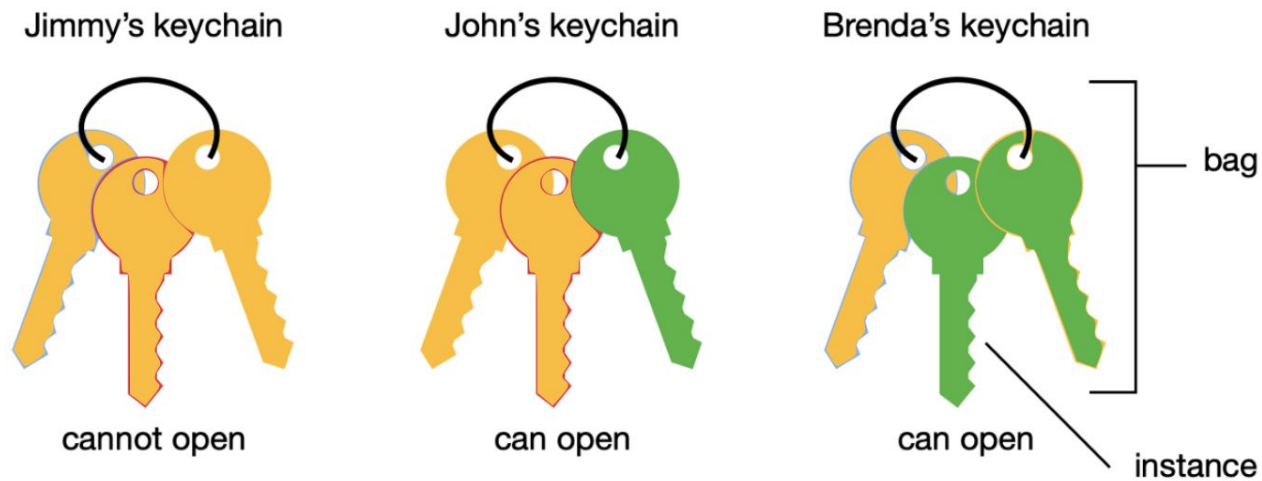
Data: Read/RNA-level



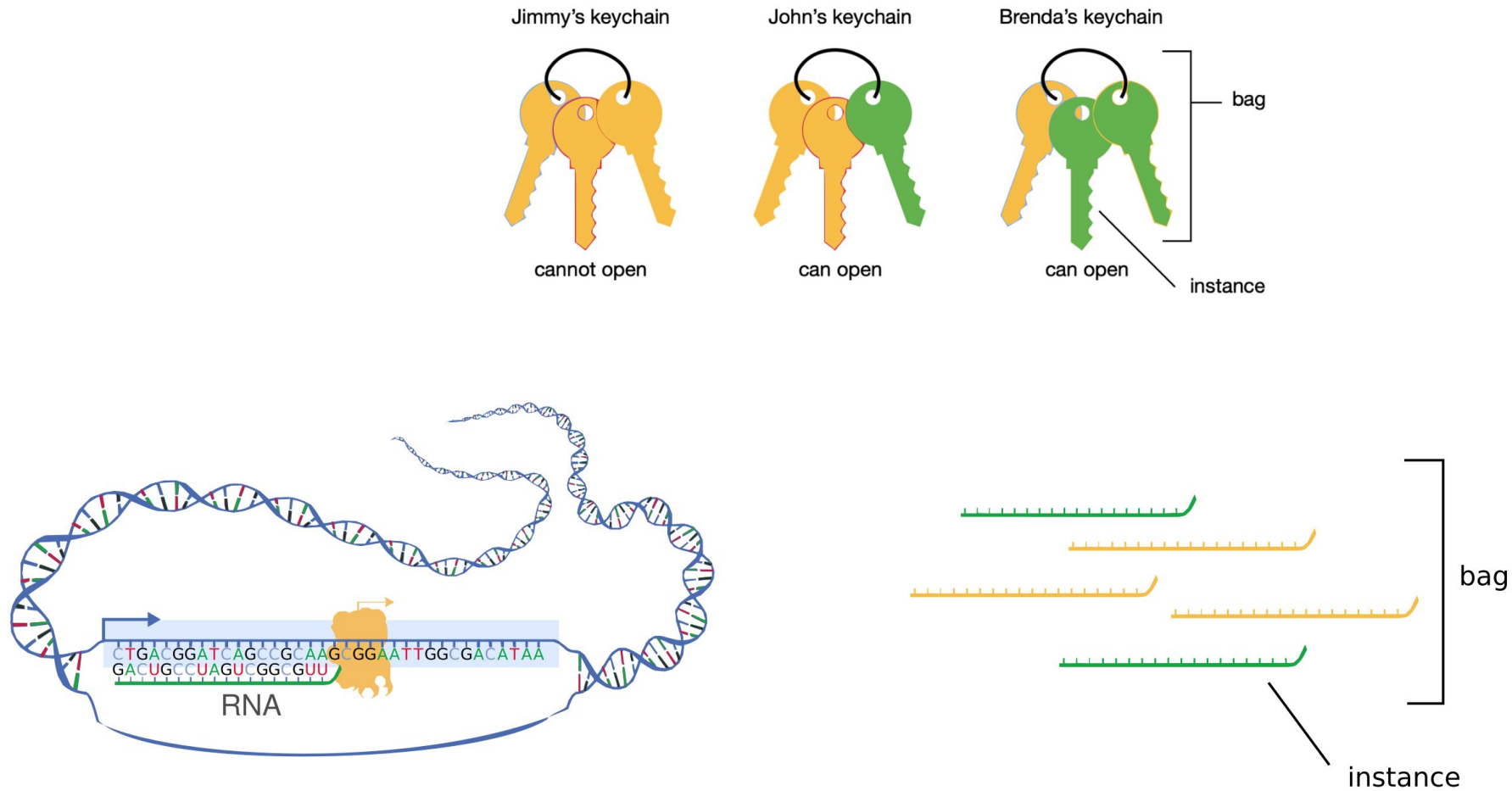
RNA



A Multiple Instance Learning (MIL) problem



A Multiple Instance Learning (MIL) problem



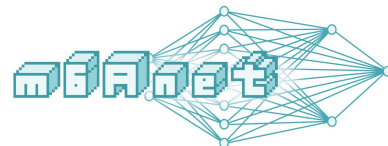
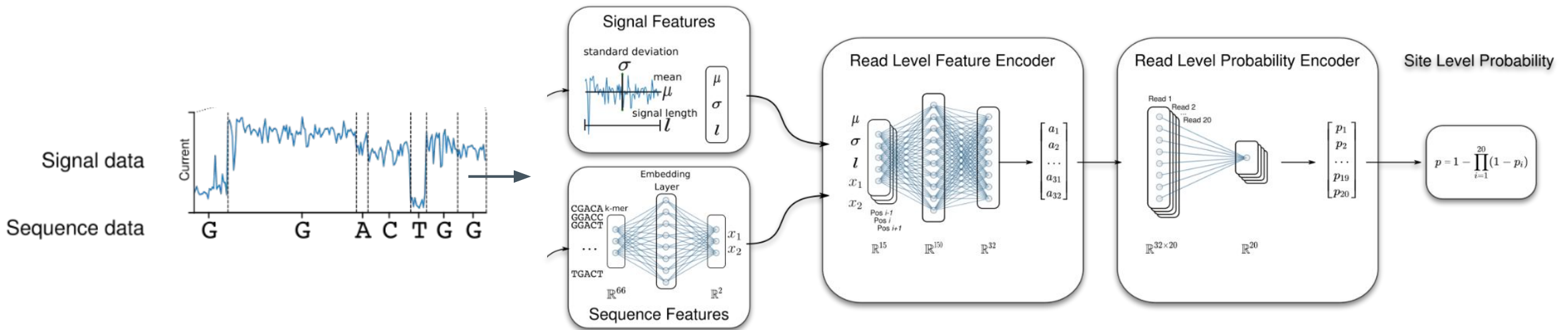
m6Anet: single sample, single molecule m6A predictions

Input

direct RNA-Seq data
1 sample

Output

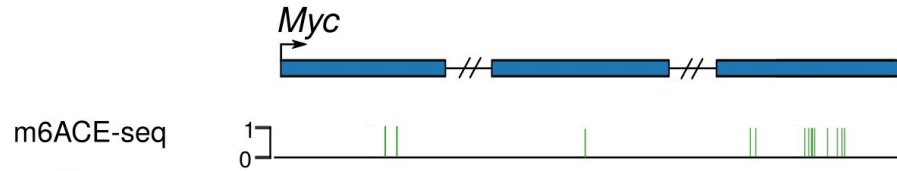
m6A modification probability
(single molecule + site)



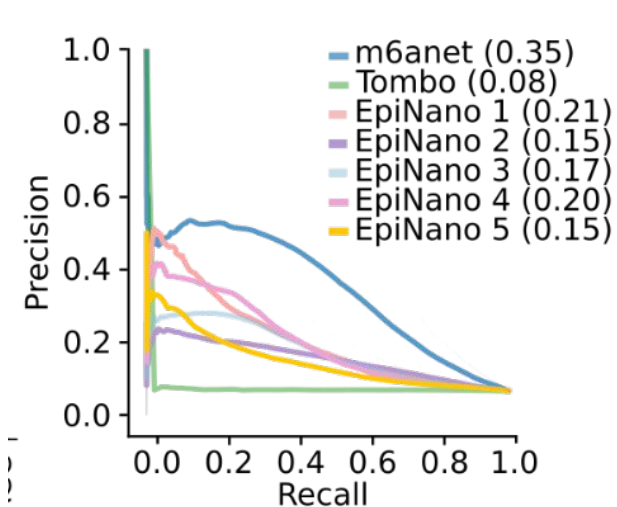
Christopher Hendra, Sho Goh
Hendra *et al.* (2022) *Nature Methods*
github.com/GoekeLab/m6anet



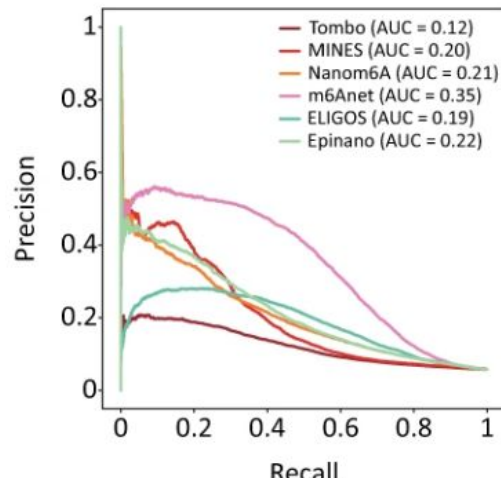
m6Anet: single sample, single molecule m6A predictions



- m6Anet accurately identifies m6A modifications from direct RNA-Seq data
- **trained on site-level data**
- **predicts site level** and **read level** modification probabilities
- m6Anet **generalises well** to new context (human, mouse, arabidopsis)



Hendra et al. *Nature Methods* (2022)



Zhong, Zhen-Dong, et al.
Nature Communications 14.1 (2023): 1906.



Christopher Hendra, Sho Goh
Hendra et al. (2022) *Nature Methods*
github.com/GoekeLab/m6anet

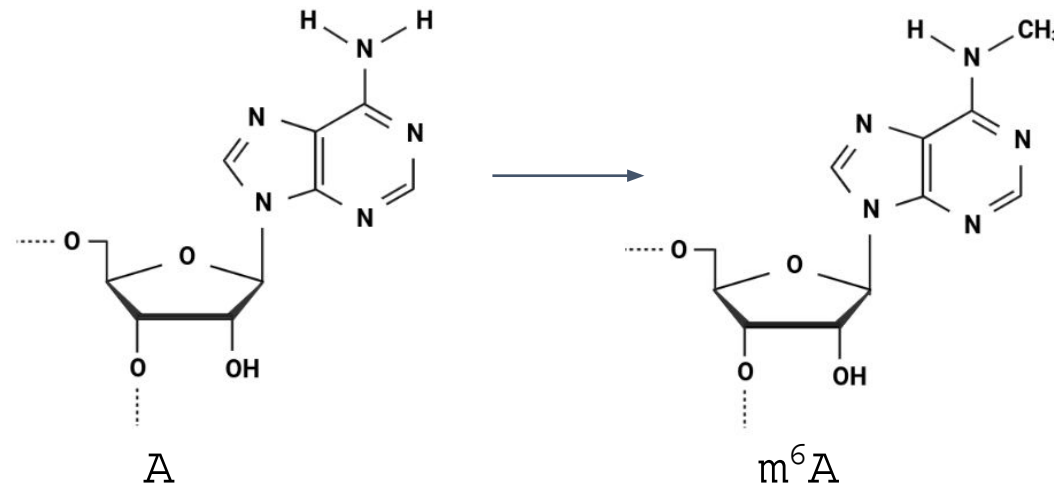
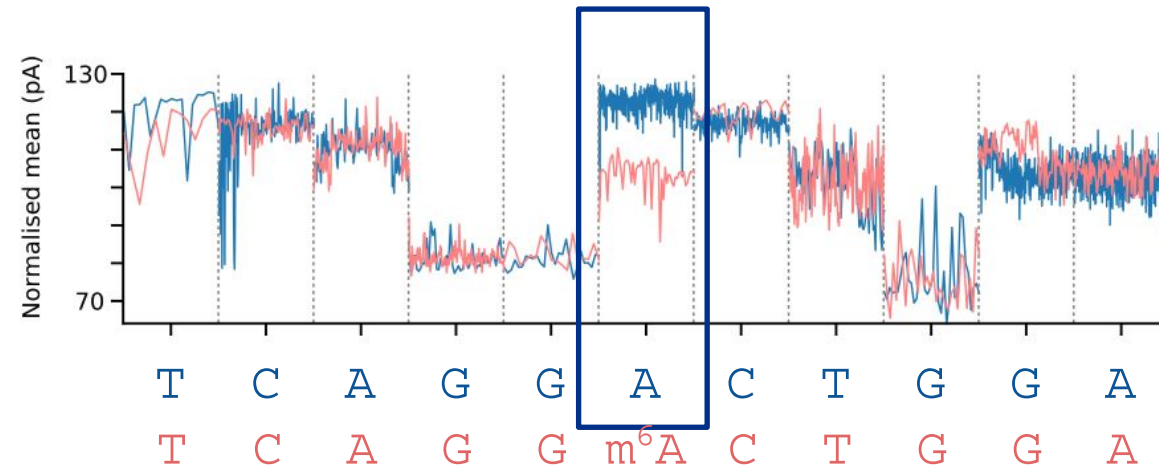
Summary

- RNA modifications are chemical alterations of the RNA bases
- RNA modifications can change the function of RNAs
 - Essential for mRNA vaccines
 - Deregulated in cancer
 - Impact almost any function of RNAs
- m6A is one of the most frequent mRNA modifications
- Nanopore direct RNA-Seq can be used to identify m6A modifications

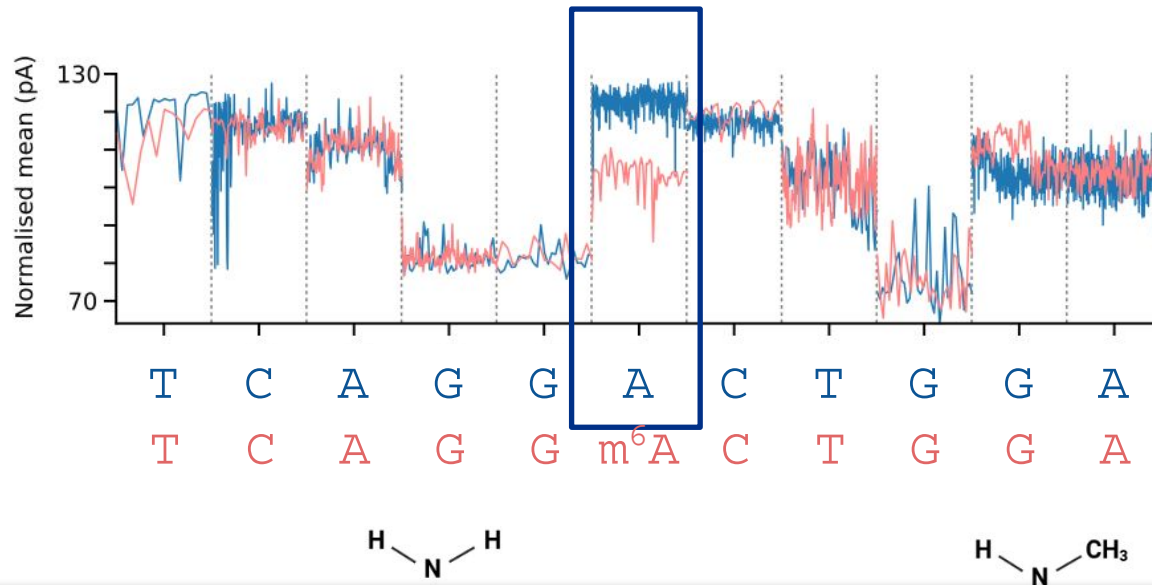
Team Project:

Identification of RNA modifications from direct RNA-Seq

Identification of RNA modifications from direct RNA-Seq



Identification of RNA modifications from direct RNA-Seq



Aim of this project:

- Develop a machine learning classifier that identifies m⁶A RNA modifications from direct RNA-Seq data
- Apply your method to identify m⁶A in the human cancer cell lines from the SG-NEx data

Project Overview

- **Task 1: Train a classifier to detect m6A modification from direct RNA sequencing data**
 - We have segmented the raw signals and extracted features from candidate m6A positions
 - Each feature represents +/-1 flanking position from the candidate m6A positions
 - Each position contains variable number of reads and one label to describe whether the given position contains m6A as given by m6ACE protocol (Koh et al 2019)
- **Task 2: Prediction of m6A sites in all SG-NEx direct RNA-Seq samples**
 - apply your method to analyse direct RNA-Seq SG-NEx data (all available cell lines)
 - demonstrate your data science expertise, domain knowledge, and **creativity** to analyse a real-world data set
- **Submission requirements and assessment components**

Project Overview: Task 1

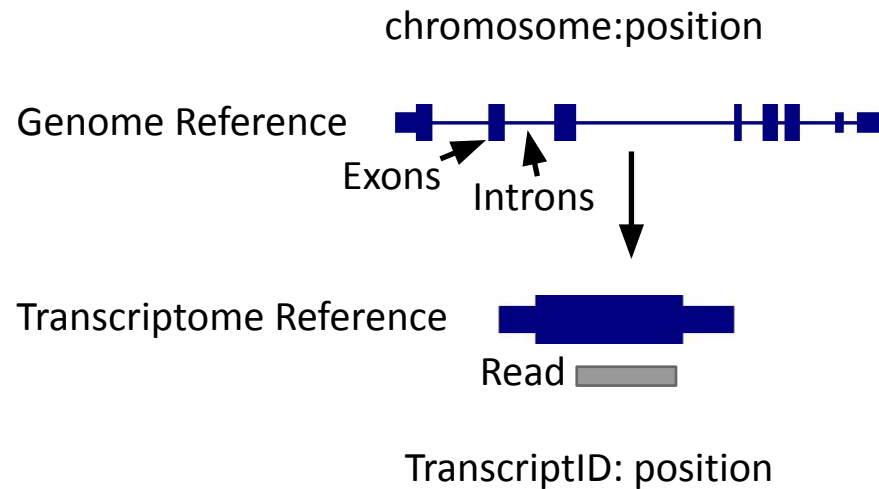
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 - apply your method to analyse direct RNA-Seq SG-NEx data (all available cell lines)
 - demonstrate your data science expertise, domain knowledge, and creativity to analyse a real-world data set
- **Submission requirements and assessment components**

Data Description

For training you are given two files:

- **data.json** contains **features**
- **data.info** contains **metadata** and **label** information

RNA-Seq: Transcriptome vs Genome Alignment



- **Genome alignment:**

- Aligning reads to the reference genome (chromosome sequences) allows the discovery of new genes and transcripts
- Genome alignment is required for BamBu
- e.g. Chr1: 159,300,450

- **Transcriptome alignment:**

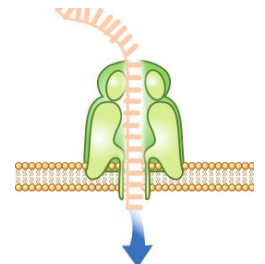
- The RNA transcript sequences can be used as a **reference transcriptome**
- transcriptome alignment is used for RNA modification data
- e.g. ENST00000135150: 249

The nanopore sequencing data: data.json

- Example from data.json:
 - ```
{ "ENST00000000233": { "244": { "AAGACCA": [[0.00299, 2.06, 125.0, 0.0177, 10.4, 122.0, 0.0093, 10.9, 84.1], [0.00631, 2.53, 125.0, 0.00844, 4.67, 126.0, 0.0103, 6.3, 80.9], [0.00465, 3.92, 109.0, 0.0136, 12.0, 124.0, 0.00498, 2.13, 79.6], [0.00398, 2.06, 125.0, 0.0083, 5.01, 130.0, 0.00498, 3.78, 80.4], [0.00664, 2.92, 120.0, 0.00266, 3.94, 129.0, 0.013, 7.15, 82.2], [0.0103, 3.83, 123.0, 0.00598, 6.45, 126.0, 0.0153, 1.09, 74.8], [0.00398, 3.75, 126.0, 0.00332, 4.3, 129.0, 0.00299, 1.93, 81.9], [0.00498, 3.93, 127.0, 0.00398, 2.51, 131.0, 0.0111, 3.47, 79.4] ...] }
```
- Each line contains reads from a single position, i.e position 244 from transcript ENST00000000233

# The nanopore sequencing data: data.json

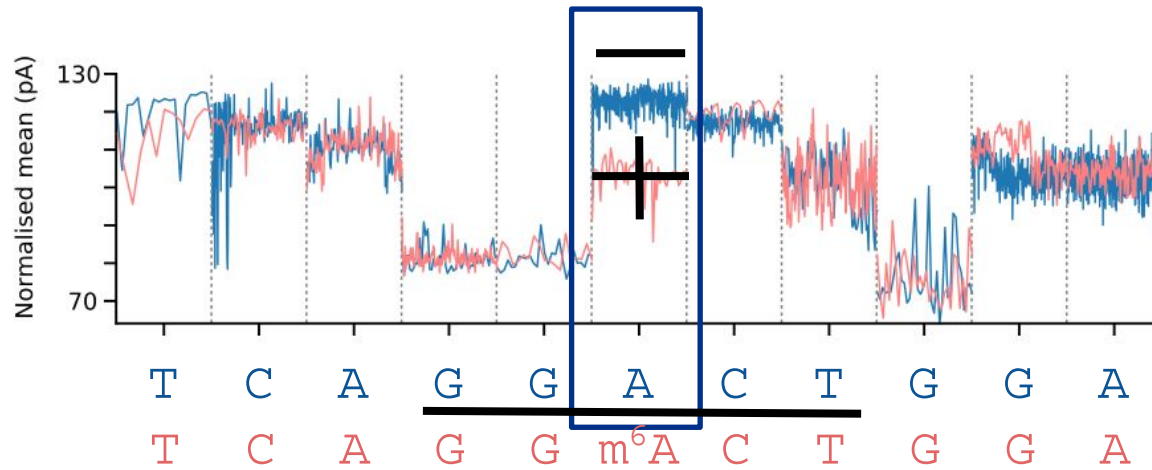
- Example from data.json:
  - ```
{ "ENST00000000233": { "244": { "AAGACCA": [ [0.00299, 2.06, 125.0, 0.0177, 10.4, 122.0, 0.0093, 10.9, 84.1], [0.00631, 2.53, 125.0, 0.00844, 4.67, 126.0, 0.0103, 6.3, 80.9], [0.00465, 3.92, 109.0, 0.0136, 12.0, 124.0, 0.00498, 2.13, 79.6], [0.00398, 2.06, 125.0, 0.0083, 5.01, 130.0, 0.00498, 3.78, 80.4], [0.00664, 2.92, 120.0, 0.00266, 3.94, 129.0, 0.013, 7.15, 82.2], [0.0103, 3.83, 123.0, 0.00598, 6.45, 126.0, 0.0153, 1.09, 74.8], [0.00398, 3.75, 126.0, 0.00332, 4.3, 129.0, 0.00299, 1.93, 81.9], [0.00498, 3.93, 127.0, 0.00398, 2.51, 131.0, 0.0111, 3.47, 79.4] ... ] }
```
- In direct RNA sequencing, we assume that a 5-mer nucleotide passes through the pore each time and it shifts one nucleotide at a time
 - So AAGACCA corresponds to 5-mer nucleotides from three positions, AAGAC, AGACC, GACCA since we have extracted +- 1 flanking position from each m6A candidate positions
 - The middle 5-mer nucleotide (AGACC in this example) is always a DRACH motif (D=A, G, or U, R= A or G while H is A, C or T)



The nanopore sequencing data: data.json

- Example from data.json:
 - ```
{ "ENST00000000233": { "244": { "AAGACCA": [[0.00299, 2.06, 125.0, 0.0177, 10.4, 122.0, 0.0093, 10.9, 84.1], [0.00631, 2.53, 125.0, 0.00844, 4.67, 126.0, 0.0103, 6.3, 80.9], [0.00465, 3.92, 109.0, 0.0136, 12.0, 124.0, 0.00498, 2.13, 79.6], [0.00398, 2.06, 125.0, 0.0083, 5.01, 130.0, 0.00498, 3.78, 80.4], [0.00664, 2.92, 120.0, 0.00266, 3.94, 129.0, 0.013, 7.15, 82.2], [0.0103, 3.83, 123.0, 0.00598, 6.45, 126.0, 0.0153, 1.09, 74.8], [0.00398, 3.75, 126.0, 0.00332, 4.3, 129.0, 0.00299, 1.93, 81.9], [0.00498, 3.93, 127.0, 0.00398, 2.51, 131.0, 0.0111, 3.47, 79.4] ...] }
```
- The nested list represent features from +- 1 flanking position of all reads from transcript ENST00000000233 at position 244
- The features in order are dwelling time, standard dev, mean current at position 243, followed by position 244, and 245 (9 features in total per read)

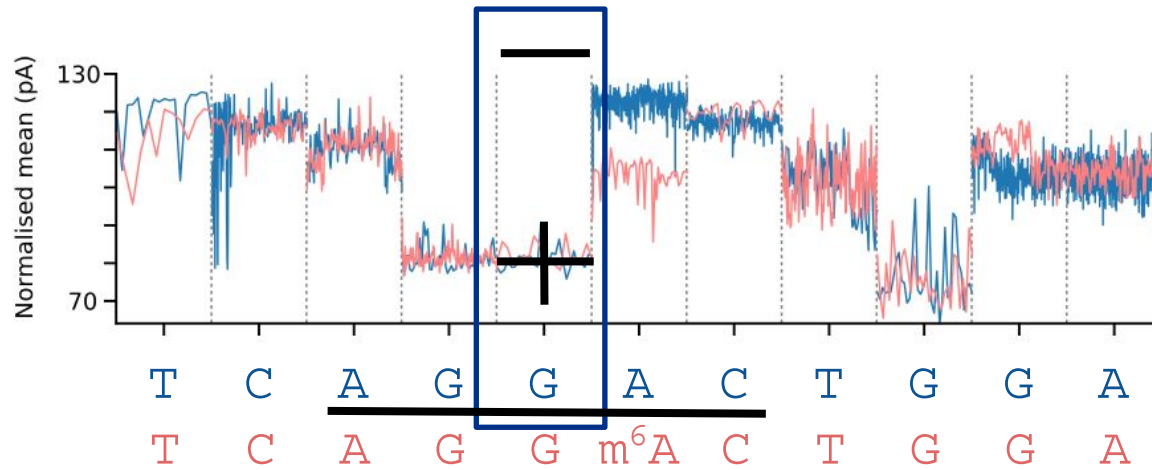
# The nanopore sequencing data: data.json



Features for each read:

- Mean
- Standard deviation
- Signal length (dwell time)
- Sequence (5-mer)
- At position 0

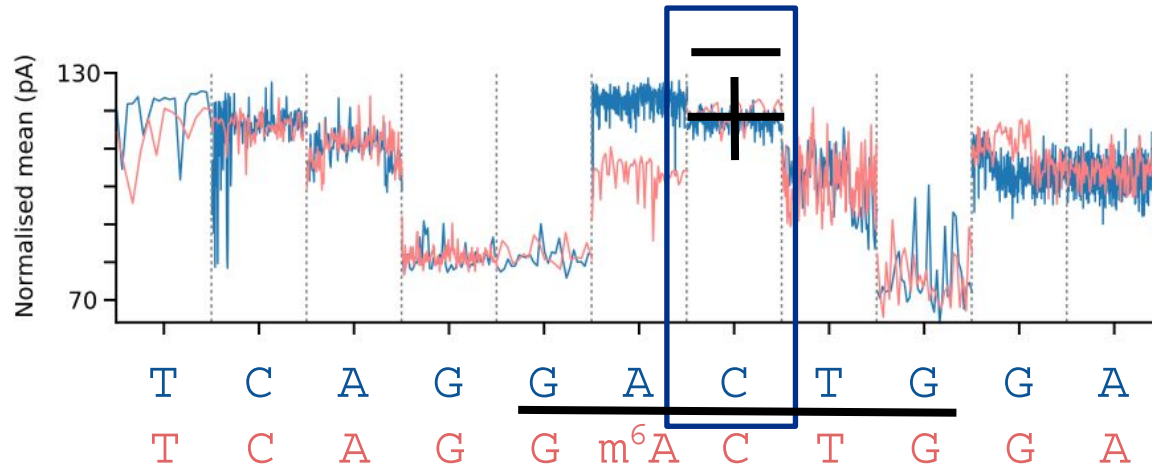
# The nanopore sequencing data: data.json



Features for each read:

- Mean
- Standard deviation
- Signal length (dwell time)
- Sequence (5-mer)
- At position 0 ,-1

# The nanopore sequencing data: data.json



Features for each read:

- Mean
- Standard deviation
- Signal length (dwell time)
- Sequence (5-mer)
- At position 0, -1, +1

# Labels: data.info

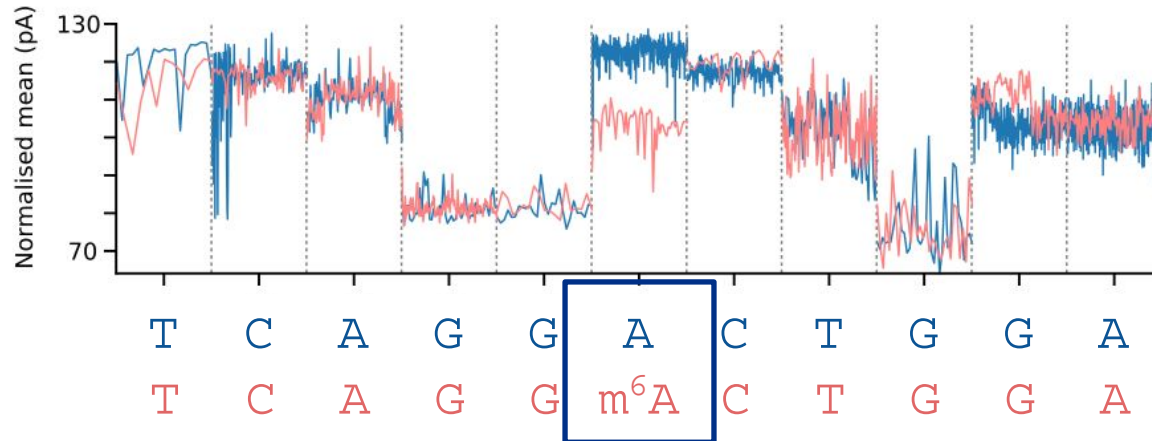
- First lines of data.info

```
gene_id,transcript_id,transcript_position,label
ENSG00000004059,ENST00000000233,244,0
ENSG00000004059,ENST00000000233,261,0
ENSG00000004059,ENST00000000233,316,0
```

- A csv file detailing the label corresponding to each transcript in data.json

- 0: unmodified; 1: modified with m6A
- The label is for the position (Transcript id + position)
- each position has a different number of reads
- we don't have the information which reads are modified at any position
- we assume that most reads at modified positions are modified

# Identification of RNA modifications from direct RNA-Seq



Features for each read:

- Mean
- Standard deviation
- Signal length (dwell time)
- Sequence (5-mer)
- At position 0, -1, +1

Labels:

- m6A positions identified by m6ACE-Seq
  - Hct116 cell line (colon cancer)

Data:

- Direct RNA-Seq
  - Hct116 cell line (colon cancer)

## Project: Output Format

- 3 columns: transcript\_id, transcript\_position, score
- Score describes the probability that the given position harbours m6A modification
- File must be comma-separated (csv format)

```
transcript_id,transcript_position,score
ENST00000005260,425,0.03794821493241728
ENST00000005260,467,0.03789510558677051
ENST00000005260,542,0.08970181109437239
ENST00000005260,585,0.014504787924707616
```

# Challenges

- **How to parse** information from json file?
- **Model selection**
  - Train-test split: by gene\_id! (think about why... try to conceptualise/visualise the data)
- **Imbalanced Classification**
  - m6A is rare, most positions do not contain m6A
  - What are some of the strategies to overcome this?

# Challenges

- **Feature extraction**

- `{"ENST00000000233":{"244":{"AAGACCA": [[0.00299,2.06,125.0,0.0177,10.4,122.0,0.0093,10.9,84.1], [0.00631,2.53,125.0,0.00844,4.67,126.0,0.0103,6.3,80.9], [0.00465,3.92,109.0,0.0136,12.0,124.0,0.00498,2.13,79.6], [0.00398,2.06,125.0,0.0083,5.01,130.0,0.00498,3.78,80.4], [0.00664,2.92,120.0,0.00266,3.94,129.0,0.013,7.15,82.2], [0.0103,3.83,123.0,0.00598,6.45,126.0,0.0153,1.09,74.8], [0.00398,3.75,126.0,0.00332,4.3,129.0,0.00299,1.93,81.9], [0.00498,3.93,127.0,0.00398,2.51,131.0,0.0111,3.47,79.4] ... ]}}`
- Each position contains variable number of reads
- Each read is described by a set of 9 features
- How do we represent the features from each position?
  - Take average for each feature across all reads? What is the limitation of this?
  - Can we do better feature extraction?

# Classifier Evaluation

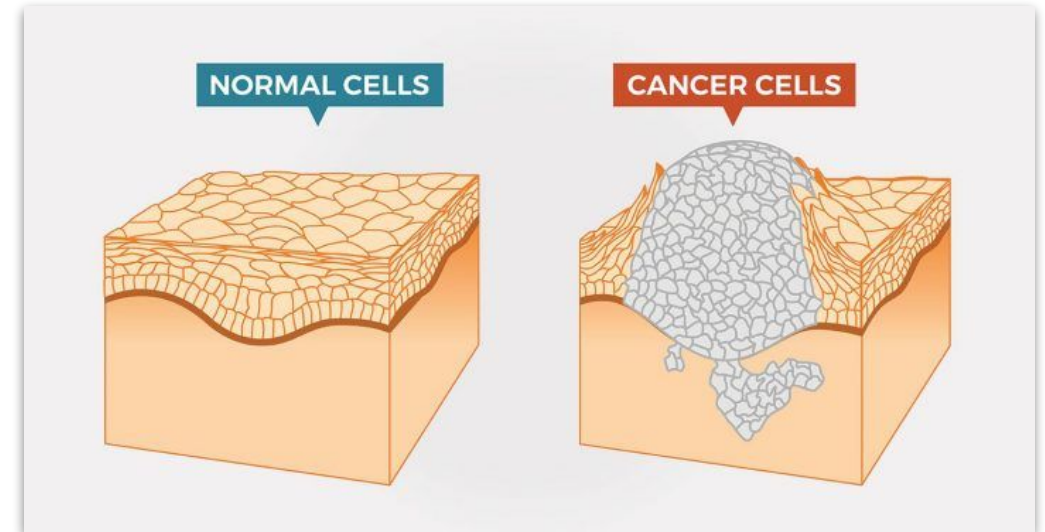
- Models will be evaluated on the ROC AUC and PR AUC
- Classification accuracy will be determined on **separate validation data (this may be a different cell type or type of data)**
- Your results will be ranked, AI can generate a 'convincing' classifier - it is your job to iterate & rationally optimise
- You will be graded on:
  - **Clarity** of communication
  - Factual **accuracy**
  - **Rigour**, critical judgement
  - Thoroughness & best practices

# Project Overview: Task 2

- **Task 1: Train a classifier to detect m6A modification from direct RNA sequencing data**
  - We have segmented the raw signals and extracted features from candidate m6A positions
  - Each feature represents +/-1 flanking position from the candidate m6A positions
  - Each position contains variable number of reads and one label to describe whether the given position contains m6A as given by m6ACE protocol (Koh et al 2019)
- **Task 2: Prediction of m6A sites in all SG-NEx direct RNA-Seq samples**
  - apply your method to analyse direct RNA-Seq SG-NEx data (all available cell lines)
  - demonstrate your data science expertise, domain knowledge, and creativity to analyse a real-world data set
- **Submission requirements and assessment components**

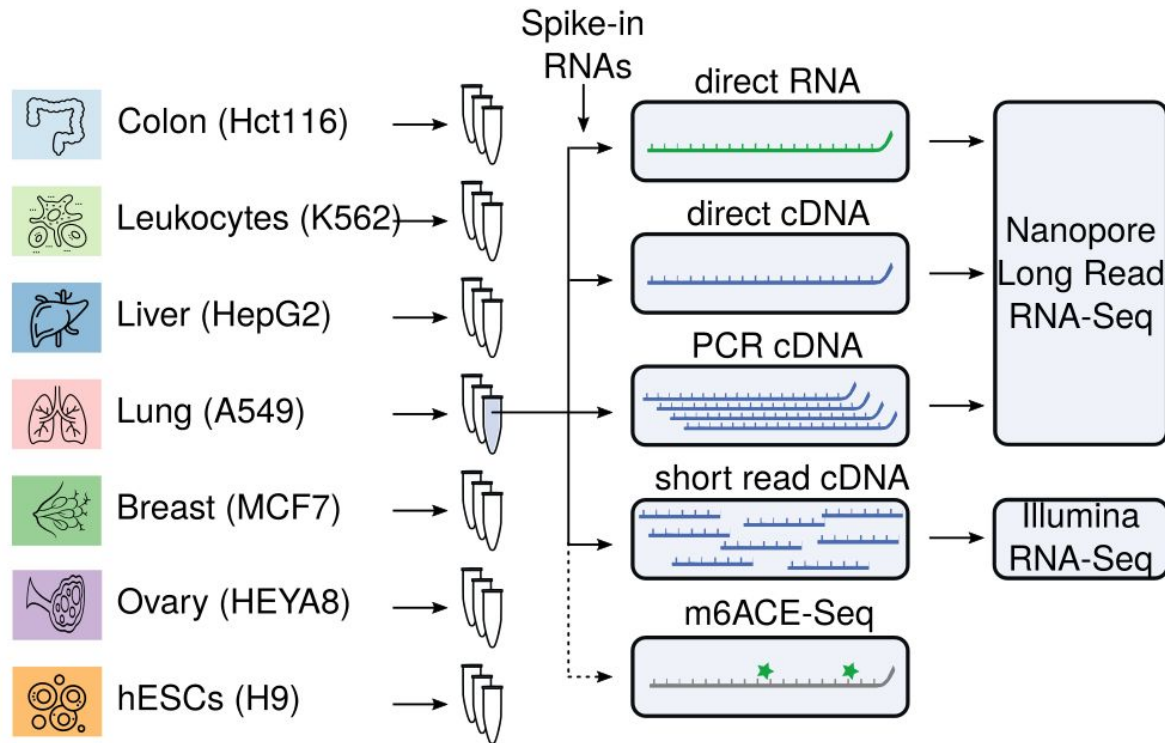
## Task 2: A analysis of RNA modifications in 5 cancer cell lines

- **Cancer cell lines** are commonly used to study cancer in the laboratory
- These cell lines are also a **model system** to study gene expression and RNA modification differences across tissues



<https://www.cancer.gov/about-cancer/understanding/what-is-cancer>; Image: Biorender

# The Singapore Nanopore Expression Project (SG-NEx)



- The SG-NEx project has generated long read RNA sequencing data that is available on AWS
- The SG-NEx data contains processed direct RNA-Seq data for multiple cancer cell lines
- Data access tutorials:  
<https://github.com/GoekeLab/sg-nex-data>

## Task 2: A analysis of RNA modifications in 5 cancer cell lines

- Processed direct RNA-Seq is available for multiple cancer cell lines
- **Task 2:**
  - Apply your method to these cell lines
  - Describe the results, and compare RNA modifications across the different cell lines. Summarise and visualise your observations. The task is intentionally open-ended.
  - Create an interactive data visualisation platform for other researchers to explore this data
- Demonstrate your **data science expertise**, **domain knowledge**, and **creativity** to analyse a **real-world data set**

# Project Overview: Submission requirements and assessment components

- **Task 1: Train a classifier to detect m6A modification from direct RNA sequencing data**
  - We have segmented the raw signals and extracted features from candidate m6A positions
  - Each feature represents  $\pm 1$  flanking position from the candidate m6A positions
  - Each position contains variable number of reads and one label to describe whether the given position contains m6A as given by m6ACE protocol (Koh et al 2019)
- **Task 2: Prediction of m6A sites in all SG-NEx direct RNA-Seq samples**
  - apply your method to analyse direct RNA-Seq SG-NEx data (all available cell lines)
  - demonstrate your data science expertise, domain knowledge, and creativity to analyse a real-world data set
- **Submission requirements and assessment components**

# Submission requirements: 1/2

- **Intermediate Leaderboard**

- There will be an intermediate submission and leaderboard
- We will provide **new data** (json format) and evaluate your predictions on these data. Additional information will be made available with the evaluation data release.
- **Submissions are due: Oct 7th, 23.59pm.** No questions will be answered between Oct 5th 9pm and Oct 7th, please ask any questions during the class on October 5th.

- **Final Leaderboard**

- For the final submission we will provide **new data** (json format) and evaluate your predictions on these data. Additional information will be made available with the evaluation data release.
- **Submissions are due: Oct 28th, 23.59pm.** No questions will be answered between Oct 26th 9pm and Oct 28th, please ask any questions during the class on Oct 26th.

*Please refer to the Team Project handout on  
Canvas for complete details and requirements*

## Submission requirements: 2/2

- Report (~5 pages) + interactive visualisation
- Reproducible code and documentation (GitHub)
- Teamwork summary
- Student evaluation
  - each student will be asked to review a component of the other teams' projects

*Please refer to the Team Project handout on  
Canvas for complete details and requirements*

# Assessment Components

- **Report & interactive data visualisation** (~5 pages summary of problem statement, method, results, and discussion) (50%)
- **Code ML tool** needs to be **executable** and results need to be **reproducible** (documentation)(5%)
- **Model performance:** results will be evaluated using an unknown test dataset and compared against several baseline models (with ROC AUC and Precision-Recall AUC) (5%)
  - the ranking in comparison with other teams will not be used for grading
- **Individual component** - confidential peer reports, participation in student evaluation, AWS usage (5%)
  - Each student is expected to attend in person and actively contribute to the project solution
  - Each student should have a specific contribution in report writing, model development, and implementation
- AWS overspending will lead to a deduction by 5%
- **Participation** is required, no-shows will also lead to a deduction
- **The project will be evaluated based on the clarity and presentation, the domain understanding, the methodology, the innovativeness and creativity**

## Examples of 'not so good'

- The project will be evaluated based on the **clarity** and presentation, the domain understanding, the methodology, the innovativeness and creativity

**Per-read contrasts, computed before aggregation.** m6A perturbs the central 5-mer far more than the flanking ones, so the flanks of the *same read* act as an internal control for that read's own signal scaling. We add six contrasts per read (central minus each flank and minus their mean, the same for signal sd, and log dwell-time ratios); dwell times, being strongly right-skewed, are modelled on a  $\log_{10}$  scale.

**Order statistics.** Each of the 15 per-read quantities is summarised over the bag by mean, sd, the 5th/25th/50th/75th/95th percentiles, min and max — turning a variable-size, unordered bag into a fixed, permutation-invariant vector that preserves the shape of its distribution, **which is the information a multiple-instance pooling layer would otherwise have to learn.**

## Examples of 'good'

- Evaluating tools with **rigour and critical judgement**
  - Do you fully understand the dataset (& limitations of the dataset) you're working with? If not -> how can you explore & better understand the dataset?
  - Do you fully understand your model? it's limitations? Have you communicated this clearly?

# AWS cloud compute

- Every team gets US\$ 100 per student
- The AWS credits need to be sufficient for the team project
- The budget will only be managed at the team level, therefore any team member can use up all AWS credits for the team.
- We will be using Ronin (starting by next class)

## 4 Hackathon sessions

For each hackathon each teams need to create a shared google document, share it in a Zulip chanel with Jonathan & Clare:

1. the specific aims for this hackathon, and who will be working on this,

After the hackathon (by 22:00):

1. what was achieved
2. what will be done by each team member until the next hackathon



# Final Session

- The final leaderboard results will be revealed in the last lecture
- The top 3 performing teams will win a prize (worth >\$100)
- Each team member (top 3 teams) will get a certificate



# Today

- Create a private zulip channel
  - #proj-teamname
  - Add Jonathan Goeke, Clare Robinson
- Get to know your group
- Ask any questions
- Let us know about any issues
- *You can also ask us about assignment 1 - we will not answer questions related to assignment 1 after today's class!*

| Name                   | Team Name       |
|------------------------|-----------------|
| Goh Kai Swee Vincent   | Big Study Group |
| Goh Yu Siang Ranier    | Big Study Group |
| Lye Tse Yong           | Big Study Group |
| Nguyen Cao Duy         | Big Study Group |
| Sokun Thydaphea        | datahelix       |
| Tan Eng Shuen          | datahelix       |
| Tan Jia Wei            | datahelix       |
| Tan Xin Yu             | datahelix       |
| Hamzah Bin Azmi        | Dolly           |
| Joshua Han Rui Chou    | Dolly           |
| Muhammad Umar Bin M    | Dolly           |
| Nitish Kumar Sahoo     | Dolly           |
| Joshua Jules Gladiston | foursight       |
| Robin Ghosh            | foursight       |
| Siah Jin Thau          | foursight       |
| CRESCENT LIEW QI E     | foursight       |
| Jinkyung Jeon          | Genomies        |
| Melissa Teo Yi Ying    | Genomies        |
| Son Seongyeon          | Genomies        |
| CHUNG JIA CHENG        | Genomies        |

| Name                   | Team Name      |
|------------------------|----------------|
| Wang Qingyang          | Genomix        |
| Yue Xinyu              | Genomix        |
| Meredith Tan           | haribo         |
| Soo Kui Xun Benjamin   | haribo         |
| Chee Jia Yun           | LIFO           |
| Clarice Ng Kai Li      | LIFO           |
| Elgin Tan              | LIFO           |
| Tasha Lim Shi Ying     | LIFO           |
| Samuel Goh Jun Yang    | Lifo           |
| Alan Anand             | Sonic_hedgehog |
| Bryan Kee Tze Ren      | Sonic_hedgehog |
| Rajvinder Dheepak Sing | Sonic_hedgehog |
| MIHIR SINHA            | sonic_hedgehog |
| Chaemin Lee            | yellow         |
| Xue Jiarui             | yellow         |
| Tan Yit Shane          | yellow         |
| Zhang Yaoyuan          | yellow         |
| ZOEBELLE ACOSTA O      | yellow         |

**Have fun, read a lot and discuss with your team members!  
Good luck!**





# THANK YOU

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[github.com/goekelab](https://github.com/goekelab)



[@jonathangoeke](https://www.linkedin.com/in/jonathangoeke)



[@jonathangoeke](https://twitter.com/jonathangoeke)



[jonathangoeke.bsky.social](https://bsky.app/profile/jonathangoeke.bsky.social)